

The Alaoglu–Erdős Integer-Exponent Conjecture

Polynomial Approximation, Integer-Valued Interpolation,
and the Arithmetic Unit Barrier

A nonduplicative continuation of the supplied unified report

Executive verdict. No unconditional proof of the conjecture is obtained, and no published resolution was found in the literature checked through August 21, 2026. The continuation does, however, extract a new approximation-theoretic core with several exact results. The sharp Bernstein-ellipse optimization shows that a classical blockwise polynomial approximation can have enough nodes only for the smallest hypothetical output $M = 2^x = 3$; the optimal block ratio is $3 + 2\sqrt{2}$, and that entire degree-versus-node geometry cannot reach any $M \geq 5$ before coefficient arithmetic is considered. A dyadic Newton-interpolation certificate then gives short exact proofs that M^ϑ is nonintegral for $M = 3, 5, 6, 7$, where $\vartheta = \log_2 3$. A discrete minimax theorem identifies the real best error as a cofactor quotient and shows exactly how clearing its denominator raises the residual to a nonzero integer. Finally, a q -Newton/Gaussian-binomial construction gives canonical integer-valued interpolants on geometric progressions and an exact extrapolation formula; its first omitted value grows rather than shrinks. These results isolate the remaining task: construct Bernstein-quality approximants in the sparse ring $\text{Int}(\langle 2, 3 \rangle, \mathbb{Z})$, or find an equivalent source of nonarchimedean cancellation.

August 21, 2026

Abstract

Write $\vartheta = \log 3 / \log 2$. The Alaoglu–Erdős conjecture asserts that $2^x, 3^x \in \mathbb{Z}$ for real x implies $x \in \mathbb{Z}$; equivalently, an integer M with $M^\vartheta \in \mathbb{Z}$ must be a power of two. The supplied unified report already develops machine-checked structural reductions, exact conditional semigroup descriptions, determinant methods, sparse interpolation, finite verification, and the reduction to the four exponentials frontier. This continuation does not repeat that material. It asks instead whether approximation theory can convert the many conditional integral values into a subunit error and hence an exact contradiction.

The first result is an exact archimedean optimization. If a hypothetical counterexample has $M = 2^\beta$, then the 3-smooth nodes in $[X, cX]$ have asymptotic density $\frac{\log c}{\log 2 \log 3} \log X$, while best degree- d polynomial approximation to t^β on that block requires $d \sim \beta \log X / \log \rho(c)$, where $\rho(c) = (\sqrt{c} + 1)/(\sqrt{c} - 1)$. The efficiency $\log c \log \rho(c)$ has the exact maximum $2 \log^2(1 + \sqrt{2})$, attained at $c = 3 + 2\sqrt{2}$. Consequently this scheme can beat the node threshold only when $M < \exp(2 \log^2(1 + \sqrt{2}) / \log 3) = 4.113124586 \dots$; among non-powers of two, only $M = 3$ remains. The second result explains why even that case is not settled by the real minimax polynomial: normalization to a compact interval introduces a denominator cost of order X^d , overwhelming the geometric gain. Markov resolution gives the same diagnosis in a different language: $O(\log X)$ arithmetic nodes are too coarse by a factor of $\log X$ to norm degree $O(\log X)$ polynomials.

On the arithmetic side, this continuation proves a dyadic interpolation certificate. If the rational polynomial interpolating T^ϑ at $d + 1$ powers of two comes within the reciprocal of its reduced denominator at an integer M , then M^ϑ is not an integer. Quadratic instances give exact exclusions for $M = 3, 5, 6, 7$. A general discrete Chebyshev duality theorem expresses the best real uniform error on $d + 2$ nodes as $|\det V|$ divided by the sum of the adjacent cofactors. At arithmetic nodes, clearing the minimax denominator produces an integer-coefficient alternant whose error is the nonzero integer $\det V$; this is the exact “unit barrier.” A separate q -Newton theorem constructs integer-valued interpolation polynomials on $\{q^n\}$, with Gaussian-binomial basis and first extrapolation error $\prod_{i=0}^d (r - q^i)$. The formula makes clear why one-axis arithmetic interpolation is plentiful but cross-axis control is absent.

The report closes by formulating precise sufficient conditions in the ring of integer-valued polynomials on the 3-smooth semigroup, by connecting them with Bhargava p -orderings and generalized factorials, and by auditing rational, Padé, Hermite–Padé, orthogonal-polynomial, and probabilistic-rounding approaches. Recent work on matrix coefficients, exponential-product transcendence, discrete Chebyshev systems, and integer-valued polynomial bases sharpens the available language but does not remove the rank-two four-exponentials obstruction.

Contents

1	Executive answer, scope, and claim ledger	1
1.1	What this continuation does not repeat	1
1.2	Claim ledger	1
1.3	The new bottom line	2
2	Minimal setup and the approximation dictionary	3
2.1	Two dual normalizations	3
2.2	The locking principle	3
2.3	Subtracting the integer part does not help	4

3	Recent literature and the relevant frontier	5
3.1	Four exponentials and matrix coefficients	5
3.2	Hermite–Padé and explicit transcendence measures	5
3.3	Discrete Chebyshev systems and integer-valued polynomials	5
3.4	Integer Chebyshev and potential theory	6
4	Bernstein ellipses and the exact block-efficiency theorem	6
4.1	Best approximation on a multiplicative block	6
4.2	How many arithmetic nodes are available?	7
4.3	The necessary degree-versus-node inequality	7
4.4	Exact optimization	8
4.5	Interpretation	9
5	Why the real minimax polynomial does not become arithmetic	9
5.1	The normalization and denominator tax	9
5.2	Capacity on the original scale	10
5.3	The Markov resolution deficit	10
5.4	Rounding coefficients is not a small perturbation	11
6	Dyadic interpolation as an exact proof-producing certificate	12
6.1	The certificate theorem	12
6.2	Two quadratic interpolants	12
6.3	What the certificate adds, and what it does not	14
6.4	A rational-interpolation variant	14
7	Discrete minimax duality and the exact unit barrier	14
7.1	The cofactor alternant	15
7.2	Clearing the minimax denominator	15
7.3	A three-node example	16
7.4	Smith normal form and the true arithmetic optimization	16
8	q-Newton interpolation, Gaussian binomials, and the geometric-axis trap	17
8.1	The normalized geometric Newton basis	17
8.2	Termination and the entire Newton shadow	18
8.3	First extrapolation grows superexponentially	19

8.4	The cross-axis criterion	19
9	Integer-valued polynomials on the 3-smooth semigroup	20
9.1	Universal congruence polynomials	20
9.2	Bhargava orderings and generalized factorials	20
9.3	The exact arithmetic Bernstein criterion	21
9.4	A concrete p -ordering computation program	21
9.5	Why regular bases and p -adic Fourier theory matter	22
10	Rational approximation and the moving denominator threshold	22
10.1	An exact rational locking theorem	22
10.2	Why ordinary rational convergence is not the right metric	23
10.3	Padé approximants and the coefficient-field obstruction	23
11	Hermite–Padé auxiliary functions and the four-exponentials wall	24
11.1	The algebraic-point mismatch	24
11.2	Exponential polynomials with integral values	24
11.3	The polynomial-on-a-curve version	25
11.4	What the recent transcendence papers do supply	25
12	Orthogonal polynomials, Christoffel functions, and lattice rounding	26
12.1	A Gram-determinant identity	26
12.2	The arithmetic gap in quadratic norm	26
12.3	Why independent coefficient rounding is too noisy	27
13	A Müntz-space formulation and two exact approximation floors	27
13.1	The arithmetic Müntz lattice	28
13.2	A continuous-versus-arithmetic dichotomy	28
13.3	Signed exponents and the denominator price	28
14	Computational reconnaissance and reproducible certificates	29
14.1	Counting 3-smooth nodes in the optimal block	29
14.2	Exact certificate generator	29
14.3	A finite p -ordering experiment	30
14.4	Numerical trust boundary	31

15 Exact sufficient conditions extracted from approximation theory	31
15.1 The generator-three integer Bernstein target	31
15.2 A generalized-factorial norm target	31
15.3 A cofactor-denominator saving	32
15.4 A cross-axis Newton target	32
15.5 A weighted rational target	32
15.6 A new logarithmic auxiliary-function theorem	32
16 Formalization blueprint	32
16.1 Low-cost exact modules	33
16.2 Medium-cost analytic modules	33
16.3 Research-level arithmetic modules	33
17 Comparison of approximation approaches	34
18 Conclusions	35
A Detailed calculus for the optimal block	35
B Proof details for the generalized power zero bound	36
C Derivation of the discrete cofactor formula	37
D The geometric Newton identity	37
E Literature chronology and relevance matrix	38

1 Executive answer, scope, and claim ledger

1.1 What this continuation does not repeat

The supplied unified report is already a book-length synthesis. It contains the reduction to four exponentials, the exact conditional free-monoid structure of all solutions, a power-primitive odd generator, block counting and equidistribution, arbitrary-node and semigroup determinants, p -adic valuation budgets, HCIZ bounds, sparse power-curve interpolation, matrix-coefficient reformulations, boundary-dilation defects, finite scans, and a detailed Lean inventory. Repeating those arguments would obscure rather than extend them.

This document uses only the following inherited facts.

1. A hypothetical noninteger solution is irrational and in fact transcendental.
2. If β is such a solution and

$$M = 2^\beta, \quad N = 3^\beta,$$

then M, N are positive integers and

$$(2^a 3^b)^\beta = M^a N^b \in \mathbb{Z} \quad (a, b \in \mathbb{N}_0).$$

3. The functions $t^{\lambda_0}, \dots, t^{\lambda_s}$ with distinct real exponents form an extended complete Chebyshev system on $(0, \infty)$; a nonzero linear combination has at most s positive zeros, counted without multiplicity for the applications here.
4. The conjecture is a special 2×2 case of the four exponentials problem, and the known six exponentials theorem does not specialize far enough.

Everything else is rederived or newly developed below.

1.2 Claim ledger

Statement	Status	Role
Bernstein root asymptotic for t^β	[classical]	Determines the best possible archimedean exponential rate on a fixed multiplicative block.
Exact maximization of $\log c \log \rho(c)$	[proved in this continuation]	Shows that classical block polynomial approximation has enough nodes only for $M = 3$.
Arithmetic subunit locking on 3-smooth nodes	[proved in this continuation]	Converts an integer-valued approximant with error < 1 into too many exact zeros.
Integer-part shift has no asymptotic gain	[proved in this continuation]	Shows that replacing β by its fractional part merely changes the arithmetic threshold from 1 to $X^{-\lfloor \beta \rfloor}$.

Statement	Status	Role
Scaling/denominator tax	[proved in this continuation]	Proves that naive compact-interval normalization costs X^d when coefficients are cleared.
Dyadic interpolation certificate	[proved in this continuation]	Gives exact finite certificates from rational interpolation at powers of two.
Exclusions $M = 3, 5, 6, 7$	[proved in this continuation]	Independent symbolic consequences of the certificate theorem; much weaker in range than inherited finite scans.
Discrete minimax cofactor identity	[proved in this continuation]	Gives the exact real best error and the alternation vector on arbitrary arithmetic nodes.
Arithmetic alternant after denominator clearing	[proved in this continuation]	Identifies the residual as the nonzero integer determinant, explaining the unit barrier.
q -Newton/Gaussian-binomial interpolation	[proved in this continuation]	Constructs integer-valued interpolants on geometric progressions and gives exact extrapolation errors.
Large zero-free computational ranges	[inherited from the unified report]	Not rerun or enlarged here.
A complete proof of the conjecture	[open target]	No such proof is claimed.

1.3 The new bottom line

Approximation theory supplies an unexpectedly sharp archimedean opportunity. For the smallest hypothetical generator $M = 3$, best polynomial approximation has enough degrees of freedom to become subunit on more arithmetic nodes than its zero count permits. This is not merely a rough comparison: the optimal block and the exact threshold can be calculated.

The obstruction is arithmetic. The best real approximant has coefficients with no reason to take integral values on the conditional node set. Clearing denominators changes a small normalized error into a large integral residual. The central problem may therefore be stated without transcendence terminology:

Construct, for a nonintegral β with $2^\beta, 3^\beta \in \mathbb{Z}$, a degree- d polynomial that is integer-valued on $\langle 2, 3 \rangle$, has error < 1 from t^β on at least $d + 2$ positive nodes, and retains Bernstein-quality size.

The report proves that the real and arithmetic sides separately have the right scale in the edge case $M = 3$, but no construction currently satisfies both.

2 Minimal setup and the approximation dictionary

2.1 Two dual normalizations

Set

$$\vartheta := \frac{\log 3}{\log 2} = 1.584962500721156 \dots$$

There are two equivalent ways to organize a counterexample.

Exponent normalization. Assume that $\beta \notin \mathbb{Z}$ and

$$M = 2^\beta \in \mathbb{N}, \quad N = 3^\beta \in \mathbb{N}.$$

Then

$$\beta = \log_2 M, \quad M^\vartheta = N.$$

The fixed arithmetic node set

$$\mathcal{B} := \{2^a 3^b : a, b \in \mathbb{N}_0\}$$

satisfies $q^\beta \in \mathbb{Z}$ for every $q \in \mathcal{B}$.

Power-curve normalization. Fix ϑ and regard M as a positive integer. The conjecture is equivalent to

$$M^\vartheta \in \mathbb{Z} \implies M = 2^m \text{ for some } m \in \mathbb{N}_0.$$

The first normalization is better for approximation on the fixed node set $\mathcal{B}$; the second is better for interpolation at the known points $(2^m, 3^m)$ of the curve $y = x^\vartheta$.

2.2 The locking principle

The basic mechanism is an integer-versus-small-real comparison.

Lemma 2.1 (Chebyshev zero bound). *Let $\lambda_0 < \dots < \lambda_s$ be real. If*

$$F(t) = \sum_{j=0}^s c_j t^{\lambda_j}$$

is not identically zero, then F has at most s distinct zeros on $(0, \infty)$.

Proof. Put $t = e^u$. Then $F(e^u) = \sum_j c_j e^{\lambda_j u}$ is an exponential polynomial with strictly ordered real frequencies. Its Wronskian equals a nonzero Vandermonde factor times a positive exponential. Hence the functions form an extended complete Chebyshev system. $\square$

Theorem 2.2 (Arithmetic subunit locking). *Assume $q^\beta \in \mathbb{Z}$ for every q in a finite set $E \subset (0, \infty) \cap \mathbb{Z}$. Let $P \in \mathbb{R}[T]$ have degree at most d and satisfy $P(q) \in \mathbb{Z}$ for every $q \in E$ (for example, $P \in \text{Int}(E, \mathbb{Z})$). If*

$$\max_{q \in E} |q^\beta - P(q)| < 1,$$

then $q^\beta = P(q)$ for every $q \in E$. Consequently, if $\beta \notin \mathbb{N}_0$ and $\#E \geq d + 2$, no such P exists.

Proof. For each $q \in E$, both q^β and $P(q)$ are integers. A difference of absolute value less than one must vanish. If there are $d + 2$ zeros, the nonzero generalized polynomial $q^\beta - P(q)$ contradicts Lemma 2.1 applied to $1, t, \dots, t^d, t^\beta$. $\square$

The same argument does not require ordinary powers.

Theorem 2.3 (Quantized Müntz approximation). *Assume $M = 2^\beta, N = 3^\beta \in \mathbb{Z}$ and define*

$$\Lambda_\beta := \{n + k\beta : n, k \in \mathbb{N}_0\}.$$

For every $q \in \mathcal{B}$ and $\lambda \in \Lambda_\beta$, one has $q^\lambda \in \mathbb{Z}$. Let $\lambda_1, \dots, \lambda_s, \mu$ be distinct elements of Λ_β . For any finite $E \subset \mathcal{B}$ with $\#E \geq s + 1$,

$$\inf_{c_1, \dots, c_s \in \mathbb{Z}} \max_{q \in E} \left| q^\mu - \sum_{j=1}^s c_j q^{\lambda_j} \right| \geq 1.$$

Likewise, every nonzero integer-coefficient Müntz polynomial with s terms has supremum at least one on any s distinct nodes of $\mathcal{B}$.

Proof. Integrality follows from

$$(2^a 3^b)^{n+k\beta} = (2^a 3^b)^n (M^a N^b)^k.$$

If the displayed maximum were below one, all residuals would vanish. The residual has at most s positive zeros by Lemma 2.1, whereas E has at least $s + 1$ nodes. The homogeneous statement is identical with one fewer term. $\square$

Remark 2.4. Theorem 2.3 is the approximation-theoretic dual of an integer generalized Vandermonde determinant. The determinant language is already developed in the supplied report; the new point here is that the lower bound is exactly the arithmetic unit 1, while the corresponding real minimax error may be exponentially smaller.

2.3 Subtracting the integer part does not help

One might hope to reduce a large β to its fractional part before approximating. The arithmetic threshold changes by exactly the amount that cancels this advantage.

Lemma 2.5 (Integer-shift invariance of the degree scale). *Write $\beta = m + \delta$ with $m \in \mathbb{N}_0$ and $0 < \delta < 1$. For $q \in \mathcal{B}$ and $P \in \mathbb{Z}[T]$,*

$$q^m(q^\delta - P(q)) = q^\beta - q^m P(q) \in \mathbb{Z}.$$

Thus a sufficient pointwise locking condition is

$$|q^\delta - P(q)| < q^{-m}.$$

On a block $[X, cX]$, best degree- d real polynomial approximation to t^δ has archimedean scale $X^\delta \rho(c)^{-d}$, so reaching the threshold X^{-m} still requires

$$d \geq \frac{\beta \log X}{\log \rho(c)} + o(\log X).$$

Proof. The integrality identity is immediate. The approximation statement follows from the Bernstein root asymptotic proved in Section 4: the condition $X^\delta \rho(c)^{-d} < X^{-m+o(1)}$ is equivalent to the displayed degree bound. $\square$

This simple cancellation is a recurring warning: lowering the analytic exponent usually introduces an exactly compensating denominator.

3 Recent literature and the relevant frontier

3.1 Four exponentials and matrix coefficients

Waldschmidt’s 2023 survey places the question “ $2^t, 3^t \in \mathbb{Z}$ implies $t \in \mathbb{N}_0$?” explicitly inside the four exponentials problem and records that both remain open [4]. Dasgupta and Kakde subsequently formulated the Matrix Coefficient Conjecture for matrices of logarithms of algebraic numbers and proved a substantial converse direction linking it with four exponentials and structural rank [5, 6]. The supplied unified report already specializes that framework to the present 2×2 matrix. Approximation theory does not bypass this frontier automatically: its coefficient lattice is another form of the same rank-collapse problem.

Massold’s 2025 preprint gives new lower bounds for transcendence degrees of fields generated by exponentials of products and reduces a stronger estimate to a torus-intersection conjecture [7]. The parameter range does not supply the missing 2×2 case here. The importance of the work is diagnostic: even contemporary advances naturally stop at a geometric intersection statement when the product matrix is critical.

3.2 Hermite–Padé and explicit transcendence measures

Fischler and Rivoal’s 2025 preprint develops a new explicit transcendence measure for values of the exponential function. The construction optimizes an accessory parameter in a Siegel determinant built from Hermite–Padé approximants to powers of $\exp$ [8]. This is very close in spirit to what one would like here: an auxiliary function with unusually small archimedean value and controlled arithmetic coefficients.

The hypotheses, however, are materially different. Classical E -function and Hermite–Padé machinery obtains arithmetic information from differential equations over algebraic function fields and evaluation at algebraic points. In the present problem the natural functions are

$$z \mapsto e^{z \log 2}, \quad z \mapsto e^{z \log 3},$$

whose differential coefficients $\log 2, \log 3$ are transcendental, and the hypothetical evaluation point $z = \beta$ is transcendental. Faverjon and Adamczewski’s recent algebraic independence measures for E - and M -functions likewise emphasize algebraic systems and algebraic evaluation points [9]. Section 11 turns this mismatch into an explicit failure audit.

3.3 Discrete Chebyshev systems and integer-valued polynomials

Gorbachev, Ivanov, and Tikhonov developed a discrete Chebyshev alternance theorem and a Remez-type algorithm for restrictions of Chebyshev systems to ordered discrete sets [10]. Their framework

is directly relevant to the cofactor minimax identity in Section 7: the conditional arithmetic nodes form a strictly ordered discrete set and the power functions form a Chebyshev system.

On the arithmetic side, Berger and Sprang connected regular bases of integer-valued polynomial rings with p -adic Fourier theory [13]. Lagarias and Yangjit’s 2025 work on B -orderings extends Bhargava’s ordering machinery for generalized factorials [12, 11]. These developments do not solve the Alaoglu–Erdős problem, but they provide exactly the language needed for the open construction in $\text{Int}(\mathcal{B}, \mathbb{Z})$.

3.4 Integer Chebyshev and potential theory

The integer Chebyshev problem studies small polynomials with integer coefficients on compact sets, with logarithmic capacity governing the first approximation scale. The monic variant was developed in detail by Borwein, Pinner, and Pritsker [15]; Pritsker’s work surveys the interaction of capacity, algebraic integers, and integer coefficients [14]. Our problem is inhomogeneous and the relevant ring is integer-valued on a sparse semigroup rather than $\mathbb{Z}[T]$. Nevertheless, potential theory explains why normalization to $[1, c]$ is archimedeanly optimal and arithmetically dangerous.

Status: T

he literature review above is [classical]. No source located through August 21, 2026 claims a proof of the Alaoglu–Erdős conjecture or of the general four exponentials conjecture. The new theorems below are independent paper proofs and are not claimed to be machine checked.

4 Bernstein ellipses and the exact block-efficiency theorem

4.1 Best approximation on a multiplicative block

For $c > 1$, $\alpha > 0$ with $\alpha \notin \mathbb{N}_0$, and $d \in \mathbb{N}_0$, define

$$E_d(\alpha; c) := \inf_{P \in \mathbb{R}[T], \deg P \leq d} \max_{1 \leq t \leq c} |t^\alpha - P(t)|.$$

The affine map from $[1, c]$ to $[-1, 1]$ sends the branch point $t = 0$ to

$$s_0 = -\frac{c+1}{c-1} < -1.$$

The Bernstein ellipse through s_0 has parameter

$$\rho(c) = |s_0| + \sqrt{s_0^2 - 1} = \frac{\sqrt{c} + 1}{\sqrt{c} - 1}. \quad (1)$$

Theorem 4.1 (Bernstein root asymptotic). *For fixed $c > 1$ and positive nonintegral α ,*

$$\lim_{d \rightarrow \infty} E_d(\alpha; c)^{1/d} = \rho(c)^{-1}.$$

Consequently, on $[X, cX]$,

$$\inf_{\deg P \leq d} \max_{X \leq t \leq cX} |t^\alpha - P(t)| = X^\alpha E_d(\alpha; c),$$

and subunit real error requires

$$d \geq \frac{\alpha \log X}{\log \rho(c)} + o(\log X). \quad (2)$$

Proof. The function t^α , with the principal branch, is holomorphic in the slit plane $\mathbb{C} \setminus (-\infty, 0]$. Under the affine map, it is holomorphic inside every Bernstein ellipse of parameter smaller than $\rho(c)$, and the branch point at 0 prevents analytic continuation through the ellipse of parameter $\rho(c)$. Bernstein’s converse theorem for geometric polynomial approximation gives the root asymptotic. The scaling identity follows by writing $t = Xu$ and absorbing X^α into the coefficients. $\square$

4.2 How many arithmetic nodes are available?

Let

$$B(X, c) := \#(\mathcal{B} \cap [X, cX]).$$

Put $a = \log 2$, $b = \log 3$, $L = \log X$, and $h = \log c$. The condition $2^m 3^n \in [X, cX]$ is

$$L \leq am + bn \leq L + h.$$

Since a/b is irrational, the boundary phases are equidistributed.

Proposition 4.2 (Thin-strip count). *For every fixed $c > 1$,*

$$B(X, c) = \frac{\log c}{\log 2 \log 3} \log X + o(\log X) \quad (X \rightarrow \infty). \quad (3)$$

Proof sketch. Sum over $n \leq (L + h)/b$. For each n , the admissible m lie in an interval of fixed length h/a whose endpoints have fractional parts obtained from the irrational rotation $n(b/a)$. Weyl equidistribution gives average occupancy h/a ; the number of n is $L/b + O(1)$. The resulting main term is $hL/(ab)$ and the discrepancy is $o(L)$. $\square$

The supplied report proves stronger blockwise equidistribution statements conditionally for candidate semigroups. Proposition 4.2 is the fixed-node analogue needed here.

4.3 The necessary degree-versus-node inequality

Assume a counterexample β and suppose, optimistically, that one can find a degree- d polynomial integer-valued on $\mathcal{B}$ whose error from t^β is below one throughout $[X, cX]$. Theorem 2.2 needs $B(X, c) \geq d + 2$. Comparing (2) and (3) gives the leading-order necessary inequality

$$\frac{\log c}{\log 2 \log 3} \geq \frac{\beta}{\log \rho(c)}. \quad (4)$$

Because $M = 2^\beta$, this is

$$\boxed{\log c \log \rho(c) \geq \log 3 \log M.} \quad (5)$$

A strict inequality is needed for a positive asymptotic degree margin; equality would require lower-order analysis.

The left side depends only on block geometry. It can be optimized exactly.

4.4 Exact optimization

Theorem 4.3 (Optimal multiplicative block). *For $c > 1$, set*

$$\Psi(c) := \log c \log \frac{\sqrt{c} + 1}{\sqrt{c} - 1}.$$

Then

$$\max_{c>1} \Psi(c) = 2 \log^2(1 + \sqrt{2}), \quad (6)$$

and the unique maximizer is

$$c_* = 3 + 2\sqrt{2} = (1 + \sqrt{2})^2. \quad (7)$$

At this point $\rho(c_) = 1 + \sqrt{2}$.*

Proof. Write

$$y = \frac{1}{2} \log c, \quad u = \log \rho(c) = \log \coth(y/2).$$

The identity $\tanh(y/2) = e^{-u}$ gives

$$\sinh y \sinh u = 1.$$

We claim that $yu \leq \operatorname{arsinh}(1)^2$. The function

$$z \mapsto \log \sinh(e^z)$$

is strictly convex, because with $x = e^z$ its second derivative is

$$x \coth x - x^2 \operatorname{csch}^2 x = \frac{x(\sinh x \cosh x - x)}{\sinh^2 x} > 0.$$

Applying midpoint convexity at $\log y$ and $\log u$ yields

$$\sinh \sqrt{yu} \leq \sqrt{\sinh y \sinh u} = 1.$$

Hence $\sqrt{yu} \leq \operatorname{arsinh}(1) = \log(1 + \sqrt{2})$. Equality forces $y = u$, so $\sinh^2 y = 1$, whence $y = u = \operatorname{arsinh}(1)$. Finally $\Psi(c) = 2yu$, giving (6), and $c = e^{2y} = (1 + \sqrt{2})^2$. $\square$

Corollary 4.4 (The generator-three frontier). *A blockwise Bernstein-rate polynomial strategy can satisfy the leading-order degree-versus-node inequality only if*

$$M \leq \exp\left(\frac{2 \log^2(1 + \sqrt{2})}{\log 3}\right) = 4.113124586019410 \dots \quad (8)$$

A positive asymptotic margin requires the strict inequality. Among positive integers that are not powers of two, only $M = 3$ qualifies.

Proof. Combine (5) with Theorem 4.3. The numerical value follows by direct evaluation. The only integer M with $2 < M < 4.113 \dots$ is 3; $M = 4$ corresponds to the integral exponent $\beta = 2$. $\square$

M	$\beta = \log_2 M$	node coefficient	degree coefficient	margin
3	1.5849625	2.3148354	1.7982868	+0.5165486
4	2.0000000	2.3148354	2.2691853	+0.0456501
5	2.3219281	2.3148354	2.6344426	−0.3196072
6	2.5849625	2.3148354	2.9328795	−0.6180441
7	2.8073549	2.3148354	3.1852043	−0.8703689
9	3.1699250	2.3148354	3.5965736	−1.2817382

Table 2: Asymptotic coefficients of $\log X$ at the optimal block $c_* = 3 + 2\sqrt{2}$. The node coefficient is $\log c_*/(\log 2 \log 3)$; the degree coefficient is $\beta/\log(1 + \sqrt{2})$. $M = 4$ is shown only as a calibration.

4.5 Interpretation

Corollary 4.4 is both encouraging and restrictive.

- It is encouraging because the archimedean side is not uniformly too weak. For $M = 3$ it has a genuine positive asymptotic margin.
- It is restrictive because no sharpening of constants inside ordinary polynomial approximation can make this particular block-count method reach $M \geq 5$. A different function space, a different arithmetic source of nodes, or a nonarchimedean gain is necessary.
- It identifies the exact place where an integer-valued approximation theorem would have the highest leverage: the ring $\text{Int}(\mathcal{B}, \mathbb{Z})$, exponent $\beta = \vartheta$, and block ratio $3 + 2\sqrt{2}$.

The next section explains why the real minimax polynomial cannot simply be rounded or scaled into that ring.

5 Why the real minimax polynomial does not become arithmetic

5.1 The normalization and denominator tax

A best approximant on $[1, c]$ has real coefficients. To use Theorem 2.2, one must turn it into a polynomial taking integer values at the arithmetic nodes. The most obvious conversion is fatal.

Proposition 5.1 (Naive coefficient clearing). *Assume $X \in \mathcal{B}$ and put $A = X^\beta \in \mathbb{Z}$. Let*

$$p(u) = \sum_{j=0}^d a_j u^j \in \mathbb{Q}[u], \quad Da_j \in \mathbb{Z}$$

for a positive integer D . Define

$$\tilde{P}_X(t) := DX^d A p(t/X) \in \mathbb{Z}[t].$$

Then

$$\max_{X \leq t \leq cX} |DX^d t^\beta - \tilde{P}_X(t)| = DX^{d+\beta} \max_{1 \leq u \leq c} |u^\beta - p(u)|. \quad (9)$$

If p has Bernstein-rate error $\rho(c)^{-d+o(d)}$, the logarithm of the right side is

$$d \log X - d \log \rho(c) + \beta \log X + \log D + o(d).$$

For every choice $d \asymp \log X$, the positive term $d \log X \asymp (\log X)^2$ overwhelms the Bernstein gain $d \log \rho(c) \asymp \log X$.

Proof. The coefficient of t^j in $DX^d Ap(t/X)$ is $DAa_j X^{d-j} \in \mathbb{Z}$. The identity (9) follows after setting $t = Xu$. The asymptotic comparison is immediate. $\square$

Caution 5.2. Proposition 5.1 is a no-go theorem only for the naive operation “approximate on $[1, c]$, rescale, clear all coefficient denominators.” It does not rule out polynomials that are integer-valued on $\mathcal{B}$ without having integer coefficients. That larger ring is the principal open direction in Section 9.

The same tax appears if one normalizes several 3-smooth nodes by a common node X . Ratios such as $2^a 3^b / X$ have numerator and denominator of order X when the exponent vectors move in opposite directions. A common coefficient denominator for a degree- d polynomial in those ratios can therefore be of order X^d . The obstruction is not numerical roundoff; it is the arithmetic cost of changing scale.

5.2 Capacity on the original scale

Potential theory gives a complementary formulation.

Proposition 5.3 (Monic Chebyshev norm on a block). *Let $I = [X, cX]$ and $d \geq 1$. Among monic real polynomials of degree d ,*

$$\min_{P \text{ monic}} \|P\|_I = 2 \left(\frac{(c-1)X}{4} \right)^d. \quad (10)$$

In particular,

$$\text{cap}(I) = \frac{(c-1)X}{4}.$$

Proof. Scale the monic Chebyshev polynomial on $[-1, 1]$. Since T_d has leading coefficient 2^{d-1} and uniform norm one, the monic extremal polynomial on an interval of length $L = (c-1)X$ has norm $L^d / 2^{2d-1} = 2(L/4)^d$. $\square$

On the normalized interval $[1, c]$, capacity is fixed and geometric approximation is possible. On the original interval, capacity grows linearly with X . Integer coefficients live naturally on the original scale, whereas complex approximation lives naturally on the normalized scale. Proposition 5.1 is the coefficient-level expression of this conflict.

The classical integer Chebyshev problem asks for a nonzero $P \in \mathbb{Z}[T]$ with small norm on a fixed compact set. Here one asks for an inhomogeneous approximation to t^β , and the right arithmetic class is $\text{Int}(\mathcal{B}, \mathbb{Z})$. Capacity remains an indispensable lower-order constraint, but it cannot alone see the sparse congruences of $\mathcal{B}$.

5.3 The Markov resolution deficit

Could one control a polynomial on the full block from its values on the arithmetic nodes and then use a real minimax construction? Standard norming arguments fail by one logarithm.

For a finite set $E \subset I = [X, cX]$, define its fill distance

$$h(E, I) := \max_{t \in I} \min_{q \in E} |t - q|.$$

Markov's inequality on an interval of length $L = (c - 1)X$ gives

$$\|P'\|_I \leq \frac{2d^2}{L} \|P\|_I \quad (\deg P \leq d).$$

Therefore

$$\|P\|_I \leq \|P\|_E + \frac{2d^2 h(E, I)}{L} \|P\|_I. \quad (11)$$

When $2d^2 h/L < 1$, the nodes are a norming mesh with constant $(1 - 2d^2 h/L)^{-1}$.

Proposition 5.4 (Arithmetic nodes are below Markov resolution). *Let $E_X = \mathcal{B} \cap [X, cX]$, with fixed $c > 1$. If $d \asymp \log X$, then*

$$\frac{2d^2 h(E_X, [X, cX])}{(c - 1)X} \rightarrow \infty$$

after passage to any sequence on which the implicit degree constant is positive. More precisely,

$$h(E_X, [X, cX]) \geq \frac{(c - 1)X}{2(B(X, c) + 1)},$$

so the Markov factor is at least $d^2/(B(X, c) + 1) \asymp \log X$.

Proof. The $B(X, c)$ nodes split the interval into at most $B(X, c) + 1$ gaps. One gap has length at least $L/(B(X, c) + 1)$, and its midpoint is at distance at least half that length from every node. Proposition 4.2 gives $B(X, c) \asymp \log X$. $\square$

A degree- d polynomial has boundary-layer resolution L/d^2 . The 3-smooth nodes have spacing scale L/d when $d \asymp \log X$. Thus a degree- d polynomial can form a narrow peak inside a typical arithmetic gap. This does not prove that every discrete approximation scheme fails, but it proves that the most common Markov-mesh transfer cannot supply the missing estimate.

A measure-theoretic Remez inequality is even less adapted: a finite node set has measure zero. Thickening each node to the Markov scale L/d^2 gives total measure only $O(B(X, c)L/d^2) = O(L/\log X)$, and the resulting Remez constant grows roughly like a power of $\log X$ raised to degree d . Its logarithm is of order $\log X \log \log X$, larger than the $O(\log X)$ Bernstein margin.

5.4 Rounding coefficients is not a small perturbation

Let $p(t) = \sum_{j=0}^d a_j t^j$ approximate t^β on $[X, cX]$. Rounding each coefficient to a nearby integer creates a pointwise perturbation bounded only by

$$\frac{1}{2} \sum_{j=0}^d (cX)^j \asymp X^d.$$

Using a Newton or Chebyshev basis reduces conditioning over the real numbers but does not by itself make the basis integer-valued on $\mathcal{B}$. The correct rounding problem is a closest vector problem in the evaluation lattice

$$\mathcal{L}_E = \{(P(q))_{q \in E} : P \in \mathbb{Z}[T], \deg P \leq d\} \subset \mathbb{Z}^E,$$

or, more flexibly, in the evaluation image of $\text{Int}(\mathcal{B}, \mathbb{Z})$. The cofactor identity of Section 7 computes the continuous optimum, but its distance to this lattice is the unresolved quantity.

6 Dyadic interpolation as an exact proof-producing certificate

The power-curve normalization supplies infinitely many exact rational interpolation data:

$$(2^n, (2^n)^\vartheta) = (2^n, 3^n), \quad n \in \mathbb{N}_0.$$

A rational interpolant through such points can be compared with an integer target at another integer M . The reciprocal of its reduced denominator is an exact exclusion threshold.

6.1 The certificate theorem

Theorem 6.1 (Dyadic interpolation certificate). *Let $0 \leq n_0 < \dots < n_d$ and let $I_{\mathbf{n}} \in \mathbb{Q}[T]$ be the unique polynomial of degree at most d satisfying*

$$I_{\mathbf{n}}(2^{n_j}) = 3^{n_j} \quad (0 \leq j \leq d).$$

Let $M \in \mathbb{N}$ be distinct from all 2^{n_j} , and write

$$I_{\mathbf{n}}(M) = \frac{a}{b}$$

in lowest terms with $b > 0$. If

$$\left| M^\vartheta - \frac{a}{b} \right| < \frac{1}{b}, \tag{12}$$

then $M^\vartheta \notin \mathbb{Z}$.

Proof. Suppose $M^\vartheta \in \mathbb{Z}$. The difference $M^\vartheta - a/b$ is rational. It cannot vanish: otherwise $T^\vartheta - I_{\mathbf{n}}(T)$ would have the $d+2$ distinct positive zeros $2^{n_0}, \dots, 2^{n_d}, M$, contradicting Lemma 2.1. A nonzero difference between an integer and a reduced fraction with denominator b has absolute value at least $1/b$, contradicting (12). $\square$

Remark 6.2. This is a genuine polynomial-approximation proof certificate, not a floating-point test. The interpolant and its denominator are exact. The only analytic input is a certified interval for M^ϑ narrow enough to verify (12). Such an interval can be proved by integer power comparisons using rational bounds for ϑ .

6.2 Two quadratic interpolants

Interpolation at 1, 2, 4 gives

$$I_{0,1,2}(T) = \frac{T^2 + 3T - 1}{3}. \tag{13}$$

Interpolation at 2, 4, 8 gives the scaled polynomial

$$I_{1,2,3}(T) = \frac{T^2 + 6T - 4}{4}. \tag{14}$$

Both identities are checked by substitution.

The following rational bounds are elementary:

$$\frac{19}{12} < \vartheta < \frac{8}{5}, \quad \vartheta < \frac{27}{17}. \quad (15)$$

Indeed,

$$3^{12} = 531441 > 524288 = 2^{19}, \quad 3^5 = 243 < 256 = 2^8,$$

and

$$3^{17} = 129140163 < 134217728 = 2^{27}.$$

Corollary 6.3 (Four short symbolic exclusions). *One has*

$$3^\vartheta, 5^\vartheta, 6^\vartheta, 7^\vartheta \notin \mathbb{Z}.$$

Equivalently, no Alaoglu–Erdős counterexample can have $2^\beta \in \{3, 5, 6, 7\}$.

Proof. For $M = 3$, (13) gives $I(3) = 17/3$. The exact comparisons

$$3^{31} = 617673396283947 > 582622237229761 = 17^{12}, \quad 3^8 = 6561 < 7776 = 6^5$$

imply

$$\frac{17}{3} < 3^{19/12} < 3^\vartheta < 3^{8/5} < 6.$$

Hence $|3^\vartheta - 17/3| < 1/3$, and Theorem 6.1 applies.

For $M = 5$, the same interpolant gives $I(5) = 13$. The integer comparisons

$$5^{19} = 19073486328125 > 8916100448256 = 12^{12}, \quad 5^8 = 390625 < 537824 = 14^5$$

give $12 < 5^\vartheta < 14$, so $|5^\vartheta - 13| < 1$.

For $M = 6$, (14) gives $I(6) = 17$. Since

$$6^{19} = 609359740010496 > 582622237229761 = 17^{12}, \quad 6^8 = 1679616 < 1889568 = 18^5,$$

we obtain $17 < 6^\vartheta < 18$.

For $M = 7$, (14) gives $I(7) = 87/4$. The lower comparison

$$7^{19} 4^{12} = 191461105130195933708288 > 188031682201497672618241 = 87^{12}$$

and the upper comparisons in (15), together with

$$7^{27} = 65712362363534280139543 < 8771994066994086373138432 = 22^{17},$$

give

$$\frac{87}{4} < 7^{19/12} < 7^\vartheta < 7^{27/17} < 22.$$

Thus $|7^\vartheta - 87/4| < 1/4$. □

M	interpolation nodes	$I(M)$	certified interval for M^ϑ	threshold
3	1, 2, 4	17/3	(17/3, 6)	1/3
5	1, 2, 4	13	(12, 14)	1
6	2, 4, 8	17	(17, 18)	1
7	2, 4, 8	87/4	(87/4, 22)	1/4

Table 3: Exact dyadic interpolation certificates. The open interval has width at most twice the threshold around the displayed interpolant value; equality with the interpolant is excluded by the Chebyshev zero bound.

6.3 What the certificate adds, and what it does not

The inherited finite verification in the supplied report excludes a vastly larger range of candidate generators. Corollary 6.3 is therefore not a record computation. Its value is conceptual:

- it is a short symbolic proof based entirely on polynomial approximation and exact integer inequalities;
- it makes the denominator threshold explicit;
- it produces independently checkable certificates consisting only of a node list, a rational polynomial value, and a rational interval for M^ϑ ;
- it generalizes immediately to any set of dyadic interpolation nodes.

For larger M , the interpolation error and the reduced denominator compete. Moving the nodes near M improves real approximation but introduces powers of two into the denominator. This is a finite version of Proposition 5.1.

6.4 A rational-interpolation variant

Let $P, Q \in \mathbb{Z}[T]$ be coprime and suppose P/Q interpolates T^ϑ at $d + e + 1$ dyadic nodes, with $\deg P \leq d$ and $\deg Q \leq e$. If $Q(M) \neq 0$ and

$$|M^\vartheta Q(M) - P(M)| < 1, \quad (16)$$

then $M^\vartheta \notin \mathbb{Z}$. Indeed, an integral target would make the residual a nonzero integer. It is nonzero because $T^\vartheta Q(T) - P(T)$ contains at most $d + e + 2$ distinct powers and therefore at most $d + e + 1$ positive zeros, while equality at M would add one zero beyond the interpolation count. Section 10 develops the asymptotic version and explains why the factor $Q(M)$ is usually prohibitive.

7 Discrete minimax duality and the exact unit barrier

Real best approximation on a finite node set has a closed determinant formula. Its arithmetic interpretation reveals the missing denominator saving.

7.1 The cofactor alternant

Let $q_0 < \dots < q_N$ be positive nodes and let $\lambda_0 < \dots < \lambda_{N-1} < \mu$. Put

$$A = (q_i^{\lambda_j})_{\substack{0 \leq i \leq N \\ 0 \leq j < N}}, \quad f = (q_i^\mu)_{0 \leq i \leq N}, \quad V = [A \ f].$$

For $0 \leq i \leq N$, let

$$C_i = (-1)^{i+N} \det A_{\hat{i}},$$

where $A_{\hat{i}}$ is obtained by deleting row i . Set

$$D = \det V, \quad S = \sum_{i=0}^N |C_i|.$$

Strict total positivity of the generalized Vandermonde matrix implies $D > 0$ and that the C_i alternate in sign after a fixed global normalization.

Theorem 7.1 (Exact discrete minimax error). *With the notation above,*

$$\min_{c \in \mathbb{R}^N} \|f - Ac\|_\infty = \frac{|D|}{S}. \quad (17)$$

There is a unique minimizer, and its residual vector is

$$f - Ac_* = \frac{D}{S} (\operatorname{sgn} C_0, \dots, \operatorname{sgn} C_N)^T. \quad (18)$$

Proof. The vector $C = (C_0, \dots, C_N)^T$ spans the one-dimensional nullspace of A^T by Laplace expansion. The ℓ^∞ distance from f to $\operatorname{col}(A)$ has the dual formula

$$\min_c \|f - Ac\|_\infty = \max_{A^T y = 0, \|y\|_1 \leq 1} |y^T f|.$$

The feasible nullspace is one-dimensional, so the maximum is attained at $y = C/S$. Another Laplace expansion gives $C^T f = D$. Hence the optimum is $|D|/S$.

Let $s_i = \operatorname{sgn} C_i$. The dual equality condition for ℓ^∞ – ℓ^1 duality forces the primal residual to equal εs at every node, with $\varepsilon = D/S$. The augmented matrix $[A \ s]$ is nonsingular because its determinant is $\sum_i |C_i| = S$, so this residual and the minimizer are unique. $\square$

This is the finite-node form of Chebyshev alternation. It also appears naturally in the discrete theory of Chebyshev systems developed in [10].

7.2 Clearing the minimax denominator

Suppose now that all entries of A and f are integers, as happens conditionally on $\mathcal{B}$ for exponents in Λ_β . Then $D, C_i, S \in \mathbb{Z}$.

Theorem 7.2 (Arithmetic alternant). *Under the integral-entry hypothesis, let $s = (\operatorname{sgn} C_i)$. There exists $z \in \mathbb{Z}^N$ such that*

$$Sf - Az = Ds. \quad (19)$$

Equivalently, the integer-coefficient generalized polynomial

$$G(t) = St^\mu - \sum_{j=0}^{N-1} z_j t^{\lambda_j}$$

takes the alternating values $D \operatorname{sgn} C_i$ at the nodes q_i . In particular, the real minimax residual D/S becomes the nonzero integer residual D after the natural denominator is cleared.

Proof. Solve $Az = Sf - Ds$. By (18), the real solution is $z = Sc_*$. Since $\det[A \ s] = S$,

$$\begin{pmatrix} z \\ D \end{pmatrix} = [A \ s]^{-1} Sf = \operatorname{adj}([A \ s])f.$$

The adjugate and f are integral, so every component of z is an integer. $\square$

Remark 7.3. Theorem 7.2 is the exact arithmetic gap. The real optimum can be far below one because S is large. But the canonical scaling that makes its coefficients integral multiplies the residual by the same S , leaving $D \geq 1$. Any proof along this line must avoid paying all of S , or prove an extra divisibility of D that is incompatible with an independent upper bound.

7.3 A three-node example

Take nodes 1, 2, 3, basis 1, t , and target t^2 . Then

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad f = \begin{pmatrix} 1 \\ 4 \\ 9 \end{pmatrix}, \quad C = (1, -2, 1),$$

so $D = 2$ and $S = 4$. The best real linear approximant is

$$p_*(t) = 4t - \frac{7}{2},$$

with alternating residuals $(1/2, -1/2, 1/2)$. Clearing the cofactor denominator gives

$$4t^2 - 16t + 14,$$

whose node values are $(2, -2, 2)$. The small real error $1/2$ has become the nonzero integer 2.

For comparison, the best integer-coefficient linear approximation has uniform error one: $p(t) = 4t - 3$ gives residuals $(0, -1, 0)$. Thus the evaluation lattice may beat the canonical cofactor scaling, but it cannot cross the unit threshold on three nodes because equality at all three would violate the zero bound.

7.4 Smith normal form and the true arithmetic optimization

Let $E = \{q_0, \dots, q_N\}$ and consider the lattice $\mathcal{L} = AZ^N \subset \mathbb{Z}^{N+1}$. The exact arithmetic problem is

$$\operatorname{dist}_{\ell^\infty}(f, \mathcal{L}). \tag{20}$$

The real minimax theorem replaces $\mathcal{L}$ by its real span. Smith normal form describes the quotient $\mathbb{Z}^{N+1}/\mathcal{L}$ and its congruence obstructions, while the cofactor vector identifies the unique real dual direction. A successful argument would combine:

1. an archimedean upper bound showing that f lies within distance < 1 of $\text{span}_{\mathbb{R}} \mathcal{L}$;
2. a lattice theorem showing that the closest real point can be represented by an integral coefficient vector without increasing the error to one;
3. or, conversely, a congruence theorem forcing every nonzero residual to have absolute value much larger than the analytic upper bound.

Theorem 7.2 shows why a generic determinant estimate is not enough: D and S must be studied together.

8 q -Newton interpolation, Gaussian binomials, and the geometric-axis trap

The dyadic certificates are initial pieces of a canonical integer-valued interpolation calculus on a geometric progression. Its formulas are exact and reveal both a large arithmetic denominator budget and a matching extrapolation explosion.

8.1 The normalized geometric Newton basis

Fix integers $q \geq 2$ and $r \geq 1$. For $k \geq 0$, define

$$B_k^{(q)}(X) := \prod_{i=0}^{k-1} \frac{X - q^i}{q^k - q^i}, \quad B_0^{(q)} = 1, \quad (21)$$

and

$$a_k(q, r) := \prod_{i=0}^{k-1} (r - q^i), \quad a_0(q, r) = 1. \quad (22)$$

For $n, k \in \mathbb{N}_0$,

$$B_k^{(q)}(q^n) = \left[\begin{matrix} n \\ k \end{matrix} \right]_q := \prod_{j=0}^{k-1} \frac{q^{n-j} - 1}{q^{k-j} - 1}, \quad (23)$$

with the convention that the Gaussian binomial is zero when $k > n$. Hence the basis is integer-valued on the geometric set $\{q^n : n \geq 0\}$.

Theorem 8.1 (q -Newton interpolation). *Define*

$$P_d^{q,r}(X) := \sum_{k=0}^d a_k(q, r) B_k^{(q)}(X). \quad (24)$$

Then:

1. $P_d^{q,r}(q^n) \in \mathbb{Z}$ for every $n \in \mathbb{N}_0$;
2. $P_d^{q,r}(q^n) = r^n$ for $0 \leq n \leq d$;
3. for every $n > d$,

$$r^n - P_d^{q,r}(q^n) = \sum_{k=d+1}^n \left[\begin{matrix} n \\ k \end{matrix} \right]_q \prod_{i=0}^{k-1} (r - q^i); \quad (25)$$

4. at the first omitted geometric node,

$$r^{d+1} - P_d^{q,r}(q^{d+1}) = \prod_{i=0}^d (r - q^i). \quad (26)$$

Proof. Equation (23) follows by factoring q^i from numerator and denominator and reversing the product. Gaussian binomial coefficients are integers, proving the first claim.

The Newton interpolation formula for the function $F(q^n) = r^n$ on the nodes $1, q, q^2, \dots$ has divided-difference coefficient $\prod_{i < k} (r - q^i)$ in the normalized basis (21). Equivalently, for each n the polynomial identity in r

$$r^n = \sum_{k=0}^n \left[\begin{matrix} n \\ k \end{matrix} \right]_q \prod_{i=0}^{k-1} (r - q^i) \quad (27)$$

holds because both sides have degree n and agree at $r = 1, q, \dots, q^n$ by triangularity. Truncating (27) proves the second and third claims. For $n = d + 1$, the remainder has a single term and the Gaussian coefficient is one. $\square$

8.2 Termination and the entire Newton shadow

Let $\alpha = \log_q r$. If $r = q^m$ with $m \in \mathbb{N}_0$, then $a_{m+1}(q, r) = 0$ and all subsequent coefficients vanish. The series terminates at $P_m^{q,r}(X) = X^m$. Conversely, if r is not a nonnegative integral power of q , no coefficient vanishes.

The denominator of $B_k^{(q)}$ is

$$D_k(q) := \prod_{i=0}^{k-1} (q^k - q^i) = q^{k^2} \prod_{j=1}^k (1 - q^{-j}). \quad (28)$$

For fixed q, r with no terminating coefficient,

$$\log |a_k(q, r)| = \frac{k(k-1)}{2} \log q + O_{q,r}(1), \quad \log D_k(q) = k^2 \log q + O_q(1). \quad (29)$$

For fixed X , the k th term of (24) is $O_{q,r,X}(q^{-k})$. Consequently

$$H_{q,r}(X) := \sum_{k=0}^{\infty} a_k(q, r) B_k^{(q)}(X) \quad (30)$$

converges locally uniformly to an entire function satisfying $H_{q,r}(q^n) = r^n$ for every $n \geq 0$.

When $\alpha \notin \mathbb{N}_0$, this entire interpolation function is not X^α : the latter has a branch point at zero. Thus the geometric data admit a canonical entire “Newton shadow” different from the intended power function between the nodes. This is a special function version of the underdetermination inherent in one-axis interpolation.

For example,

$$H_{2,3}(3) = 5.698459082460503\dots, \quad 3^\vartheta = 5.704522494691118\dots$$

The difference is small but nonzero. The approximation is not the issue; the shadow has no known reason to share the second arithmetic axis.

8.3 First extrapolation grows superexponentially

Equation (26) has a particularly clear asymptotic consequence. For fixed q, r with $r \neq q^m$,

$$\log \left| r^{d+1} - P_d^{q,r}(q^{d+1}) \right| = \frac{d(d+1)}{2} \log q + O_{q,r}(1). \quad (31)$$

The same quadratic exponent that creates large admissible denominators also creates a large first extrapolation residual. Integer-valued interpolation on one geometric axis is therefore abundant but not close to minimax extrapolation.

For $q = 2, r = 3$, the first omitted errors are exact integers:

d	$3^{d+1} - P_d^{2,3}(2^{d+1})$	absolute value
0	2	2
1	2	2
2	-2	2
3	10	10
4	-130	130
5	3770	3770
6	-229970	229970
7	28746250	28746250
8	-7272801250	7272801250
9	3701855836250	3701855836250

Table 4: The exact first-extrapolation product $\prod_{i=0}^d (3 - 2^i)$.

8.4 The cross-axis criterion

Return to a hypothetical $M = 2^\beta \in \mathbb{Z}$. The polynomial $P_d^{2,M}(X)$ is integer-valued on every dyadic node and agrees with X^β at $1, 2, 4, \dots, 2^d$. The second condition $N = 3^\beta \in \mathbb{Z}$ would be contradicted by a single cross-axis estimate.

Criterion 8.2 (Cross-axis q -Newton certificate). *Write $P_d^{2,M}(3) = a/b$ in lowest terms. If*

$$\left| 3^\beta - \frac{a}{b} \right| < \frac{1}{b},$$

then $3^\beta \notin \mathbb{Z}$.

Proof. This is Theorem 6.1, since as a polynomial in M the value $P_d^{2,M}(3)$ is precisely the interpolant to M^β at $M = 1, 2, 4, \dots, 2^d$. $\square$

For $d = 2$,

$$P_2^{2,M}(3) = \frac{M^2 + 3M - 1}{3},$$

which is the polynomial used in Corollary 6.3. The open problem is to choose nodes or a more flexible integer-valued basis so that the cross-axis error beats the reduced denominator for every non-power M .

9 Integer-valued polynomials on the 3-smooth semigroup

The natural arithmetic class is

$$\text{Int}(\mathcal{B}, \mathbb{Z}) := \{P \in \mathbb{Q}[T] : P(q) \in \mathbb{Z} \text{ for every } q \in \mathcal{B}\}. \quad (32)$$

It is much larger than $\mathbb{Z}[T]$ and can absorb denominators through congruences rather than through coefficient clearing.

9.1 Universal congruence polynomials

For a prime $p \notin \{2, 3\}$ and $k \geq 1$, every $q \in \mathcal{B}$ is a unit modulo p^k . Euler’s theorem gives

$$q^{\varphi(p^k)} \equiv 1 \pmod{p^k}.$$

Hence

$$U_{p,k}(T) := \frac{T^{\varphi(p^k)} - 1}{p^k} \in \text{Int}(\mathcal{B}, \mathbb{Z}). \quad (33)$$

This elementary family proves that coefficient denominators need not be paid by a factor X^d at the evaluation stage. Sparse-set congruences can pay them uniformly.

The efficiency of this particular family is limited:

$$\frac{\log p^k}{\deg U_{p,k}} = \frac{k \log p}{p^{k-1}(p-1)}.$$

It decays rapidly in p and k . Products over many primes create larger denominators but also larger degrees. The relevant extremal question is global.

9.2 Bhargava orderings and generalized factorials

Bhargava’s p -orderings attach to an arbitrary subset S of a Dedekind domain a generalized factorial sequence measuring the largest denominators compatible with integer-valuedness [11]. Recent B -ordering work supplies simultaneous variants for collections of ideals [12]. For $S = \mathcal{B}$, the desired invariant is the characteristic fractional ideal

$$\mathfrak{I}_d(S) = \{a_d : \exists P(T) = \sum_{j=0}^d a_j T^j \in \text{Int}(S, \mathbb{Z})\}.$$

Equivalently, one wants a regular basis $F_0, F_1, \dots$ of $\text{Int}(S, \mathbb{Z})$ with controlled denominators, heights, and norms on multiplicative blocks.

The geometric Newton basis demonstrates the possible scale. On $S_q = \{q^n : n \geq 0\}$, the degree- k denominator $D_k(q)$ in (28) satisfies

$$\log D_k(q) = k^2 \log q + O(1).$$

Thus a sparse set can support quadratic-in-degree denominator growth. Since the naive normalization tax has logarithm $d \log X$ and the interesting regime has $d \asymp \log X$, both live on the $(\log X)^2$ scale. This is the first reason the integer-valued-polynomial route is not ruled out by Proposition 5.1.

The second reason is also a warning: for the canonical q -Newton interpolant, the Newton coefficient consumes about half of that quadratic budget, and the first extrapolation error grows on the same scale. The full semigroup $\mathcal{B}$ may provide better mixed-base congruences, but they must be optimized jointly with archimedean norm.

9.3 The exact arithmetic Bernstein criterion

The open construction can be stated as a theorem with a single missing hypothesis.

Criterion 9.1 (Arithmetic Bernstein criterion). *Let $\beta \notin \mathbb{N}_0$ satisfy $q^\beta \in \mathbb{Z}$ for all $q \in \mathcal{B}$. Suppose that for some fixed $c > 1$ there are arbitrarily large X , degrees $d = d(X)$, and polynomials $P_X \in \text{Int}(\mathcal{B}, \mathbb{Z})$ such that*

1. $\deg P_X \leq d$;
2. $\#(\mathcal{B} \cap [X, cX]) \geq d + 2$;
- 3.

$$\max_{q \in \mathcal{B} \cap [X, cX]} |q^\beta - P_X(q)| < 1.$$

Then no such β exists.

Proof. Apply Theorem 2.2 to the node set $\mathcal{B} \cap [X, cX]$. □

For $M = 3$, Theorem 4.3 says the degree and node counts leave a positive linear-in-log X margin at $c = c_*$. Thus it would be enough to prove an integer-valued analogue of Bernstein approximation with no loss in the exponential base. For $M \geq 5$, the same function space cannot work even under ideal arithmetic integerization; one needs rational or Müntz spaces, additional nodes, or a nonarchimedean amplifier.

9.4 A concrete p -ordering computation program

For each prime p , a p -ordering of $\mathcal{B}$ greedily chooses $a_0, a_1, \dots \in \mathcal{B}$ so that

$$v_p \left(\prod_{i < n} (a_n - a_i) \right)$$

is minimal at each stage. The resulting valuation is independent of the choices and gives the p -part of the generalized factorial. A finite rigorous program can:

1. enumerate 3-smooth numbers up to a certified cutoff;
2. compute greedy minima for each p up to a degree d ;
3. use periodicity of $2^a 3^b \bmod p^k$ to prove that the finite search is complete;
4. combine the local factorials into a candidate regular basis;
5. solve a linear or lattice optimization for minimal uniform error on $\mathcal{B} \cap [X, cX]$.

The key theorem to seek is not merely a formula for the generalized factorial. It is a simultaneous bound

$$\inf_{P \in \text{Int}(\mathcal{B}, \mathbb{Z}), \deg P \leq d} \max_{q \in \mathcal{B} \cap [X, cX]} |q^\beta - P(q)| \leq \exp(\beta \log X - d \log \rho(c) + o(\log X)), \quad (34)$$

with a loss smaller than the positive margin in Table 2. No such theorem is presently known.

9.5 Why regular bases and p -adic Fourier theory matter

Berger and Sprang show that regular bases of rings of integer-valued polynomials can be encoded by p -adic Fourier expansions and used to derive numerical criteria for arithmetic properties [13]. In the present setting, a regular basis adapted to $\mathcal{B}$ could do two things at once:

- make integer-valuedness automatic, replacing a hard global congruence constraint by integral basis coefficients;
- expose the archimedean norm of each basis element on $[X, cX]$, turning (34) into a weighted coefficient approximation problem.

The obstacle is that $\mathcal{B}$ is sparse in the real topology but highly recurrent in every p -adic unit group away from 2 and 3. That mixture is exactly what generalized factorials are designed to encode.

10 Rational approximation and the moving denominator threshold

Polynomial approximation is only one member of the classical hierarchy. Rational functions can place poles near the branch cut of t^β and may converge substantially faster on a fixed compact interval. In the arithmetic problem, however, the comparison scale is no longer the fixed number one. The denominator of the approximant is evaluated at the arithmetic node, so the locking threshold moves with the node.

10.1 An exact rational locking theorem

Theorem 10.1 (Rational-function locking). *Let $\alpha > 0$ be nonintegral and let $E \subset \mathbb{N}$ be a finite set such that $q^\alpha \in \mathbb{Z}$ for every $q \in E$. Let $P, Q \in \mathbb{Z}[T]$ be nonzero with*

$$\deg P \leq m, \quad \deg Q \leq n,$$

and suppose $Q(q) \neq 0$ for all $q \in E$. If

$$\left| q^\alpha - \frac{P(q)}{Q(q)} \right| < \frac{1}{|Q(q)|} \quad (q \in E), \quad (35)$$

then $|E| \leq m + n + 1$. Consequently, finding $m + n + 2$ such nodes satisfying (35) excludes the assumed integrality.

Proof. For every $q \in E$, the quantity

$$q^\alpha Q(q) - P(q)$$

is an integer. Inequality (35) forces that integer to vanish. Hence the generalized polynomial

$$F(t) = t^\alpha Q(t) - P(t)$$

has $|E|$ positive zeros. Its exponents lie in the disjoint sets $\{\alpha, \alpha + 1, \dots, \alpha + n\}$ and $\{0, 1, \dots, m\}$, which are disjoint because $\alpha \notin \mathbb{Z}$. Thus F has at most $m + n + 2$ nonzero terms and, by the generalized Descartes–Chebyshev zero theorem, at most $m + n + 1$ positive zeros. The function cannot vanish identically because the two exponent sets are disjoint. $\square$

For the conjecture one may take $\alpha = \beta$ and $E \subset \mathcal{B}$, or $\alpha = \vartheta$ and E in the conditional candidate semigroup $\{2^a M^b : a, b \geq 0\}$. The theorem is exact and does not assume uniform approximation on the whole interval.

10.2 Why ordinary rational convergence is not the right metric

On a fixed interval $[1, c]$, a type- (m, n) rational approximant may have a very small real error. After transporting it to $[X, cX]$, however, an integer polynomial denominator of degree n is typically of size X^n at the nodes. The arithmetic requirement is therefore not merely

$$\|t^\beta - R(t)\| = o(1),$$

but a weighted estimate

$$\max_{q \in E} |Q(q)| \left| q^\beta - \frac{P(q)}{Q(q)} \right| < 1. \quad (36)$$

If $n \asymp \log X$ and $|Q(q)|$ has its generic X^n scale, the required error is $\exp(-\Theta((\log X)^2))$. Fixed-block polynomial approximation supplies only $\exp(-\Theta(n))$, and even many standard rational schemes supply an exponential in the degree rather than the required exponential in $n \log X$.

This does not prove that every rational scheme fails. It identifies the missing property: the denominator must be arithmetically tiny on the selected smooth nodes while the numerator still approximates $t^\beta Q(t)$. In other words, the problem is a weighted rational approximation problem with a denominator constrained simultaneously in the real and nonarchimedean metrics.

Criterion 10.2 (Weighted rational certificate). *A counterexample is impossible if, for arbitrarily large X , there are $P_X, Q_X \in \mathbb{Z}[T]$ and a set $E_X \subset \mathcal{B} \cap [X, cX]$ such that*

1. $|E_X| > \deg P_X + \deg Q_X + 1$;
2. Q_X has no zero on E_X ;
3. the weighted bound (36) holds at every point of E_X .

10.3 Padé approximants and the coefficient-field obstruction

The local Padé approximants to $(1 + z)^\beta$ have coefficients rational in β . For algebraic β that feature is useful, and for rational β it gives classical irrationality constructions. Here a hypothetical counterexample has transcendental β . The Padé coefficients therefore lie in $\mathbb{Q}(\beta)$ rather than $\mathbb{Q}$, and their evaluation at an integer node is not an integer or even an algebraic number with controlled

denominator. Clearing the coefficient field would require a polynomial relation for β , precisely what is unavailable.

Approximating t^ϑ avoids an unknown exponent but replaces the coefficient field by $\mathbb{Q}(\vartheta)$. Since ϑ is transcendental, the same obstruction remains. A successful Padé construction must therefore be adapted to the exponential identities $2^\vartheta = 3$ and $M^\vartheta = N$, not merely to the Taylor series of a generic power.

Status: Theorem 10.1 and Criterion 10.2 are [\[proved in this continuation\]](#). The diagnosis of standard Padé coefficients is structural: it says why the usual algebraic-exponent machinery does not specialize, not that every possible rational auxiliary function is excluded.

11 Hermite–Padé auxiliary functions and the four-exponentials wall

Recent work of Fischler and Rivoal obtains a new transcendence measure for values of the exponential function at nonzero algebraic arguments by combining Hermite–Padé approximants to powers of the exponential function with a carefully optimized Siegel determinant [8]. Faverjon and Adamczewski likewise obtain explicit algebraic independence measures for values of E -functions and Mahler functions at algebraic points [9]. These are major improvements in quantitative auxiliary function technology. The present problem sits just outside their arithmetic hypotheses.

11.1 The algebraic-point mismatch

The four relevant exponentials can be written as

$$e^{\log 2} = 2, \quad e^{\log 3} = 3, \quad e^{\beta \log 2} = M, \quad e^{\beta \log 3} = N.$$

All four values are algebraic, but the arguments $\log 2, \log 3, \beta \log 2, \beta \log 3$ are not algebraic. Hermite–Padé theorems for E -functions obtain arithmetic control by evaluating a differential system with algebraic coefficients at an algebraic point. Here the natural differential systems

$$y' = \log 2 \, y, \quad y' = \log 3 \, y$$

already have transcendental coefficients, and the common evaluation point $z = \beta$ is itself transcendental under a counterexample. Neither side of the usual arithmetic input survives.

This is not a technical inconvenience. Replacing the logarithms by formal algebraic parameters and later specializing them would require proving that a nonzero formal auxiliary determinant cannot vanish at $(\log 2, \log 3)$. That is a statement about products of logarithms of algebraic numbers, exactly the structural-rank wall isolated in the supplied report and in the Matrix Coefficient program of Dasgupta–Kakde [5, 6].

11.2 Exponential polynomials with integral values

A direct auxiliary function has the form

$$F(z) = \sum_{(a,b) \in \Omega} c_{a,b} e^{(a \log 2 + b \log 3)z} = \sum_{(a,b) \in \Omega} c_{a,b} 2^{az} 3^{bz}, \quad c_{a,b} \in \mathbb{Z}. \quad (37)$$

At $z = \beta$ every term is an integer, so $F(\beta) \in \mathbb{Z}$. If one could arrange $0 < |F(\beta)| < 1$, the conjecture would follow.

High-order vanishing at zero would impose

$$\sum_{(a,b) \in \Omega} c_{a,b} (a \log 2 + b \log 3)^j = 0 \quad (0 \leq j < L). \quad (38)$$

For algebraic frequencies, Siegel’s lemma can solve analogous systems over a number field. Here the coefficients of (38) are polynomials in $\log 2$ and $\log 3$. Solving them with integer $c_{a,b}$ without treating those logarithms as independent is already a logarithmic-algebraic problem. Treating them as formally independent replaces each condition by all its binomial coefficient conditions

$$\sum_{(a,b) \in \Omega} c_{a,b} a^r b^{j-r} = 0, \quad 0 \leq r \leq j,$$

and consumes about $L^2/2$ degrees of freedom. This quadratic cost is the auxiliary-function shadow of the critical 2×2 four-exponentials geometry.

11.3 The polynomial-on-a-curve version

Writing $X = 2^z$ and $Y = 3^z$, equation (37) becomes $F(z) = Q(X, Y)$ with $Q \in \mathbb{Z}[X, Y]$. A counterexample supplies the integral curve point $(M, N) = (2^\beta, 3^\beta)$ on

$$Y = X^\vartheta.$$

The inherited sparse-interpolation theorem says that a nonzero polynomial with s monomials has at most $s - 1$ distinct positive zeros on an irrational power curve. Thus interpolation through R positive nodes requires at least $R + 1$ monomials. The supplied report already identifies the exact one-monomial saving that would prove the conjecture. Hermite–Padé language does not erase that deficit; it repackages it as the need for one more exact moment condition than the available support supplies.

11.4 What the recent transcendence papers do supply

The recent literature is nevertheless instructive in three ways.

1. Fischler–Rivoal show that accessory parameters in a Siegel determinant can improve a transcendence measure beyond the previously expected barrier. This encourages a search for an accessory parameter adapted to the two prime valuations rather than to a single Archimedean norm.
2. Faverjon–Adamczewski show how functional equations and differential systems can turn algebraic independence into explicit Liouville-type inequalities. A comparable inequality for the specialized logarithmic matrix would immediately be relevant, but no such theorem is presently known.
3. Massold’s 2025 work develops lower bounds for transcendence degrees of fields generated by exponentials of products and reduces stronger estimates to a torus-intersection problem [7]. The 2×2 specialization remains on the conjectural side; it does not furnish the missing auxiliary function here.

Status: This section is an audited transfer analysis, not a negative theorem about all Hermite–Padé methods. The exact obstruction is that the natural arguments and coefficient matrices are transcendental, while current quantitative E -function theorems require algebraic arithmetic data.

12 Orthogonal polynomials, Christoffel functions, and lattice rounding

Uniform approximation is the right norm for the one-unit locking principle, but least-squares approximation has a useful exact determinant formula. It exposes another duality between approximation and the integer interpolation determinants of the supplied report.

12.1 A Gram-determinant identity

Let $E = \{q_0, \dots, q_N\}$, let $A = (\phi_j(q_i))_{0 \leq i \leq N, 0 \leq j < N}$, and let $f = (\psi(q_i))_{0 \leq i \leq N}$. Give the nodes positive weights w_i and write $W = \text{diag}(w_0, \dots, w_N)$ and $V = [A \ f]$.

Proposition 12.1 (Least-squares determinant quotient). *If A has full column rank, then*

$$\min_{c \in \mathbb{R}^N} (f - Ac)^T W (f - Ac) = \frac{\det(V^T W V)}{\det(A^T W A)}. \quad (39)$$

Proof. The normal equations give the minimizer $c = (A^T W A)^{-1} A^T W f$. The minimum is the Schur complement

$$f^T W f - f^T W A (A^T W A)^{-1} A^T W f$$

of $A^T W A$ in $V^T W V$. The determinant formula for a Schur complement gives (39). $\square$

For polynomial bases, the denominator is a moment or Gram determinant and its reciprocal is a Christoffel-type quantity. Choosing weights to minimize the largest leverage score is the least-squares analogue of choosing Fekete or alternation nodes in the uniform norm.

12.2 The arithmetic gap in quadratic norm

Under a counterexample, choose $q_i \in \mathcal{B}$ and exponents from $\mathbb{N}_0 + \mathbb{N}_0 \beta$. Then A and f have integer entries. If the weights are integers, both determinants in (39) are integers. This looks promising, but the quotient can be much smaller than one because the denominator determinant may be enormous. Moreover, the real minimizer generally has rational rather than integral coefficients. Clearing its denominator again multiplies the residual to a nonzero integer vector.

The correct arithmetic object is the evaluation lattice

$$\mathcal{L}_E = A\mathbb{Z}^N \subset \mathbb{Z}^{N+1}. \quad (40)$$

The desired approximation asks for the distance of the integral target f from this lattice, not merely from the real subspace $A\mathbb{R}^N$:

$$\delta_\infty(E, A; f) := \min_{z \in \mathbb{Z}^N} \|f - Az\|_\infty. \quad (41)$$

When $|E| = N + 1$, Theorem 7.2 supplies the primitive dual functional. Let $C = (C_0, \dots, C_N)$ be its cofactor vector and let $g = \gcd(C_0, \dots, C_N)$. Then C/g is primitive and every residual $r = f - Az$ satisfies

$$\frac{C}{g} \cdot r = \frac{D}{g}. \quad (42)$$

The real lower bound is $|D|/S$; the integral lower bound is the integer optimization problem of minimizing $\|r\|_\infty$ subject to (42) and the torsion congruences of the Smith normal form of A .

Criterion 12.2 (Smith-coset criterion). *A polynomial-approximation proof would follow if, for some counterexample candidate and some node/basis family with more nodes than basis functions, one could prove from the Smith normal form that the coset $f + \mathcal{L}_E$ contains the zero vector while the Chebyshev zero theorem proves that it cannot. Equivalently, a contradiction arises if the arithmetic congruences force an exact interpolant of insufficient dimension.*

This criterion deliberately asks for exact membership, not a small real error. Since all residual coordinates are integers, every nonzero coset representative has ℓ^∞ norm at least one.

12.3 Why independent coefficient rounding is too noisy

Let c^* be a real minimax or least-squares coefficient vector and round each coordinate independently to an integer with mean c_j^* . At node q_i the rounding perturbation is

$$Z_i = \sum_j A_{ij} \xi_j, \quad |\xi_j| \leq 1.$$

Even under unbiased Bernoulli rounding,

$$\text{Var}(Z_i) = \sum_j A_{ij}^2 \text{Var}(\xi_j) \leq \frac{1}{4} \sum_j A_{ij}^2.$$

For monomial or Müntz bases on nodes of size X , the right side is dominated by powers of X and is vastly larger than one. Orthogonalizing the columns reduces the real condition number, but the orthogonal basis has nonintegral coefficients and destroys the evaluation lattice. Thus generic randomized rounding attacks the wrong lattice.

A potentially relevant replacement is discrepancy minimization in a basis already adapted to $\text{Int}(\mathcal{B}, \mathbb{Z})$: one would round coefficients in a regular integer-valued basis whose vectors are small on the selected block. This is exactly where generalized factorials and p -adic Fourier bases can matter.

13 A Müntz-space formulation and two exact approximation floors

The conditional exponent semigroup

$$\Lambda_\beta = \{n + k\beta : n, k \in \mathbb{N}_0\}$$

provides far more functions than the ordinary powers. Every finite collection of distinct exponents in Λ_β is a Müntz–Chebyshev system on $(0, \infty)$, and every such monomial is integral on the base semigroup $\mathcal{B}$ under a counterexample.

13.1 The arithmetic Müntz lattice

For finite $\Lambda \subset \Lambda_\beta$, define

$$\mathcal{M}_\mathbb{Z}(\Lambda) = \left\{ \sum_{\lambda \in \Lambda} c_\lambda t^\lambda : c_\lambda \in \mathbb{Z} \right\}.$$

If $q = 2^a 3^b$, then

$$q^{n+k\beta} = q^n (M^a N^b)^k \in \mathbb{Z}.$$

Hence every $F \in \mathcal{M}_\mathbb{Z}(\Lambda)$ is integer-valued on $\mathcal{B}$.

Theorem 13.1 (Unit floor for integer Müntz polynomials). *Let Λ contain s distinct exponents and let $E \subset \mathcal{B}$ have at least s points. Under a nonintegral counterexample,*

$$0 \neq F \in \mathcal{M}_\mathbb{Z}(\Lambda) \implies \max_{q \in E} |F(q)| \geq 1.$$

More generally, if $\mu \in \Lambda_\beta \setminus \Lambda$ and $|E| \geq s + 1$, then

$$\inf_{(c_\lambda) \in \mathbb{Z}^\Lambda} \max_{q \in E} \left| q^\mu - \sum_{\lambda \in \Lambda} c_\lambda q^\lambda \right| \geq 1. \quad (43)$$

Proof. All displayed values are integers. A strict bound below one would therefore force a zero at every node. A nonzero s -term Müntz polynomial has at most $s - 1$ positive zeros, while the inhomogeneous residual has at most $s + 1$ terms and at most s positive zeros. $\square$

This theorem is simple but conceptually decisive. Classical Müntz approximation permits arbitrarily small real errors in many settings; the arithmetic coefficient lattice has a hard floor exactly at one. The problem is to build a basis in which real approximation and integer-valuedness coexist before coefficient rounding.

13.2 A continuous-versus-arithmetic dichotomy

Because ordinary nonnegative integers already lie in Λ_β , the real span contains all algebraic polynomials and is dense on every compact interval. Thus lack of analytic flexibility is not the issue. The obstruction is entirely in the coefficient lattice. One may summarize it as

$$\underbrace{\text{dist}_\infty(t^\mu, \text{span}_\mathbb{R}\{t^\lambda\})}_{\text{can be very small}} \quad \text{versus} \quad \underbrace{\text{dist}_\infty(t^\mu, \mathcal{M}_\mathbb{Z}(\Lambda))}_{\text{at least 1}}.$$

The cofactor formula of Section 7 computes the first quantity on a finite alternation set; the Smith form of the evaluation matrix controls the second.

13.3 Signed exponents and the denominator price

Allowing negative exponents would create frequencies close to β through good rational approximations $n+k\beta \approx 0$ and could improve real approximation dramatically. But $q^{n+k\beta}$ then acquires denominators on $\mathcal{B}$. Clearing them reintroduces prime powers whose logarithm is proportional to the size of the

negative exponent. This is the Müntz version of the normalization tax: frequency localization and arithmetic integrality pull in opposite directions.

A promising restricted question is to allow Laurent exponents while keeping all denominators supported only at 2 and 3, then seek a compensating 2-adic and 3-adic divisor in the cofactor determinant. The supplied report’s signed-progression and valuation modules provide part of the bookkeeping, but no leading-order cancellation is currently known.

14 Computational reconnaissance and reproducible certificates

The computations in this continuation are diagnostic rather than evidence for a full proof. They are small enough to reproduce with exact integer or rational arithmetic, except for the single displayed value of the entire Newton shadow.

14.1 Counting 3-smooth nodes in the optimal block

The following table enumerates $\mathcal{B} \cap [X, c_*X]$ exactly and compares it with the leading term from Proposition 4.2. The modest oscillation around the asymptotic is visible even at moderate scales.

X	exact count	leading term	difference
10^2	15	10.6602	4.3398
10^3	19	15.9903	3.0097
10^4	25	21.3204	3.6796
10^6	36	31.9806	4.0194
10^8	48	42.6408	5.3592
10^{10}	57	53.3011	3.6989
10^{12}	68	63.9613	4.0387

Table 5: Exact 3-smooth node counts for $c_* = 3 + 2\sqrt{2}$.

The asymptotic coefficient $2.3148353905\dots$ is large enough to beat the real polynomial degree coefficient $1.7982868151\dots$ only in the $M = 3$ branch. Table 5 does not address integer-valuedness.

14.2 Exact certificate generator

The following compact script constructs a rational dyadic interpolant, evaluates it at a candidate M , and reports the reduced denominator. The analytic interval needed by Theorem 6.1 should be proved separately by integer power comparisons.

Listing 1: Exact dyadic interpolation certificate generator.

```

1 from fractions import Fraction
2
3
4 def interpolate(nodes, values):
5     """Return coefficients low-to-high of the exact interpolant."""
6     p = [Fraction(0)]
7     for i, xi in enumerate(nodes):
8         basis = [Fraction(1)]
9         denom = Fraction(1)
10        for j, xj in enumerate(nodes):

```

```

11         if i == j:
12             continue
13         out = [Fraction(0)] * (len(basis) + 1)
14         for k, a in enumerate(basis):
15             out[k] -= xj * a
16             out[k + 1] += a
17         basis = out
18         denom *= xi - xj
19         scale = Fraction(values[i], 1) / denom
20         if len(p) < len(basis):
21             p += [Fraction(0)] * (len(basis) - len(p))
22         for k, a in enumerate(basis):
23             p[k] += scale * a
24     return p
25
26
27 def evaluate(coeffs, x):
28     y = Fraction(0)
29     for a in reversed(coeffs):
30         y = y * x + a
31     return y
32
33
34 exponents = [0, 1, 2]
35 nodes = [2**n for n in exponents]
36 values = [3**n for n in exponents]
37 poly = interpolate(nodes, values)
38 for M in [3, 5, 6, 7]:
39     value = evaluate(poly, M)
40     print(M, value, value.denominator)

```

14.3 A finite p -ordering experiment

For a finite set $S_H = \{2^a 3^b : 0 \leq a, b \leq H\}$ and a prime p , one can greedily construct a p -ordering by selecting the next element x minimizing

$$v_p \left(\prod_{y \text{ already selected}} (x - y) \right).$$

Repeating this for many primes produces an empirical generalized-factorial profile. The experiment should record:

1. the greedy sequence and its valuation increments;
2. stabilization as H grows;
3. the leading coefficient denominator permitted in degree d ;
4. the sup norm on $\mathcal{B} \cap [X, c_* X]$ of the corresponding regular-basis element;
5. the ratio between that norm and the Bernstein minimax error.

The key output is not merely a large denominator. It is a basis element having a large admissible denominator and a small block norm at the same time. Geometric q -Newton bases supply the first property but fail the second by Theorem 8.1; ordinary Chebyshev polynomials supply the second but fail the first.

14.4 Numerical trust boundary

All tables except the Newton-shadow value are derived from exact integers or rational arithmetic and rounded only for presentation. No numerical calculation in this continuation is used as a premise in a proof of the conjecture. The symbolic exclusions in Corollary 6.3 reduce entirely to the displayed integer inequalities.

15 Exact sufficient conditions extracted from approximation theory

The preceding analysis yields several precise statements whose proof would settle the conjecture or a substantial branch of it. They are listed here in decreasing order of immediate plausibility.

15.1 The generator-three integer Bernstein target

Criterion 15.1 (Optimal-block integer Bernstein target). *Let $c_* = 3 + 2\sqrt{2}$. If there are arbitrarily large X and polynomials $P_X \in \text{Int}(\mathcal{B}, \mathbb{Z})$ such that*

$$\deg P_X < \#(\mathcal{B} \cap [X, c_*X]) - 1$$

and

$$\max_{q \in \mathcal{B} \cap [X, c_*X]} |q^\vartheta - P_X(q)| < 1,$$

then $3^\vartheta \notin \mathbb{Z}$.

Proof. Apply Theorem 2.2 with $\alpha = \vartheta$ and the indicated node set. □

This criterion focuses on the only nonintegral generator for which ordinary Bernstein geometry has enough asymptotic nodes. It is therefore the natural first branch for a p -ordering or integer-Chebyshev attack.

15.2 A generalized-factorial norm target

Let F_d be a degree- d regular-basis element of $\text{Int}(\mathcal{B}, \mathbb{Z})$, normalized by its leading coefficient. A sufficient quantitative theorem would be a bound of the form

$$\|F_d\|_{[X, c_*X]} \leq \exp(o(d)) \rho(c_*)^{-d} X^d \tag{44}$$

together with enough control of the lower-degree basis elements to approximate $X^\vartheta t^\vartheta$. The X^d factor in (44) is unavoidable for a monic polynomial; the generalized factorial must cancel it in the leading coefficient. Thus one needs

$$\log d!_{\mathcal{B}} \sim d \log X$$

on the chosen degree/scale relation, with the same basis retaining the Chebyshev norm profile. This is an adelic capacity problem rather than a purely real one.

15.3 A cofactor-denominator saving

In the notation of Theorem 7.1, the best real residual is D/S and the cleared arithmetic residual is D . A proof would follow from a construction in which the minimax coefficient vector is already integral, or more generally in which its denominator is strictly smaller than the natural cofactor sum and the remaining residual is subunit on one additional arithmetic node.

Criterion 15.2 (Cofactor saving). *Suppose a family of smooth nodes and integral Müntz basis functions admits a best alternating residual whose coefficients all have a common denominator $H < S$ and such that $H(D/S) < 1$. If the corresponding coefficient clearing preserves target coefficient one, then no counterexample exists.*

The last clause is essential: multiplying the entire residual by H changes the target from t^μ to Ht^μ . One needs a congruence that clears the approximating coefficients while leaving the target coefficient congruent to one.

15.4 A cross-axis Newton target

For each non-power integer M , choose dyadic nodes $2^{n_0}, \dots, 2^{n_d}$ and let $I_{\mathbf{n}}$ interpolate $(2^{n_j}, 3^{n_j})$. Theorem 6.1 reduces the conjecture to finding a node set for which

$$\left| M^\vartheta - I_{\mathbf{n}}(M) \right| < \frac{1}{\text{den}(I_{\mathbf{n}}(M))}.$$

The node set may depend on M . The standard consecutive dyadic nodes have excellent integer-valuedness but poor extrapolation. Nonconsecutive nodes, Leja-type node selection, or a mixture of dyadic and triadic axes are concrete alternatives.

15.5 A weighted rational target

Criterion 10.2 asks for rational functions whose denominator is small at smooth nodes. A natural sufficient theorem would construct Q from cyclotomic or universal-congruence factors so that $Q(q)$ has controlled prime divisibility for all $q \in \mathcal{B}$, while P/Q is a near-best rational approximation to t^β . Such a theorem would combine potential theory with the generalized factorial of $\mathcal{B}$.

15.6 A new logarithmic auxiliary-function theorem

Finally, any theorem producing an integer exponential polynomial F of the form (37) with $0 < |F(\beta)| < 1$ would solve the problem immediately. The moment analysis shows that the needed coefficient estimate must exploit the specialization $(\log 2, \log 3)$, not treat the two logs as generic independent variables. This is the point at which approximation theory meets the Matrix Coefficient Conjecture.

16 Formalization blueprint

The supplied repository already contains APIs for exponential-polynomial zero bounds, integer determinants, finite candidates, and several semigroup grids. The continuation can be formalized in layers.

16.1 Low-cost exact modules

OptimalBernsteinBlock

Define $\rho(c)$ and $\Psi(c)$; prove the identity $\sinh y \sinh u = 1$, the log-convexity inequality, and the maximizer $c_* = 3 + 2\sqrt{2}$. This is elementary real analysis once the convexity lemma is available.

DyadicInterpolationCertificate

Formalize rational Lagrange interpolation on dyadic nodes, the Chebyshev zero-count endpoint, and the denominator locking lemma. Instantiate it for $M = 3, 5, 6, 7$ using norm-num-certified integer powers.

DiscreteCofactorAlternation

Formalize the cofactor vector, $A^T C = 0$, the ℓ^1 dual formula, and $B(z, D)^T = Sf$. The arithmetic alternant is pure linear algebra over $\mathbb{Z}$ and $\mathbb{Q}$.

GeometricNewtonBasis

Define $B_k^{(q)}$, prove evaluation as a Gaussian binomial coefficient, derive the Newton coefficients a_k , and prove the first-extrapolation product.

RationalPowerLocking

Formalize Theorem 10.1 using the existing exponential-polynomial zero API.

16.2 Medium-cost analytic modules

PowerBernsteinRate

Formalize the affine map to $[-1, 1]$, Bernstein ellipses, the branch obstruction at zero, and the root asymptotic for best approximation. This likely requires a significant complex approximation library.

SmoothThinStripCount

Formalize the fixed-width count of $2^a 3^b$ in a multiplicative block using irrational rotation and discrepancy. Much of the equidistribution infrastructure already exists in the repository.

EntireNewtonShadow

Prove compact convergence of the geometric Newton series and its interpolation property. Only geometric estimates and standard uniform convergence are needed.

16.3 Research-level arithmetic modules

SmoothSemigroupPOrdering

Compute and certify local p -orderings of $\mathcal{B}$, generalized factorials, and regular-basis leading coefficients.

AdelicBlockNorm

Relate the generalized factorial to the real block norm of regular-basis elements. This is the central open approximation target.

EvaluationLatticeSmith

Compute Smith forms for selected smooth-node Müntz matrices and extract exact coset obstructions. Finite instances are certifiable even before an asymptotic theorem is found.

The recommended order is exact algebra first, then finite certified experiments, then the complex approximation asymptotic. None of the open arithmetic modules should be represented as a theorem until its denominator and norm estimates are both proved.

17 Comparison of approximation approaches

Method	Archimedean strength	Arithmetic advantage	Exact stopping point
Bernstein polynomial	Exponential in degree with exact rate $\rho(c)^{-d}$	None for the real minimax coefficients	Node geometry reaches only $M = 3$; scaling costs X^d
Integer Chebyshev	Directly controls integer coefficients	Values integral at all integer nodes	Original interval has capacity $\asymp X$; normalization loses integrality
$\text{Int}(\mathcal{B}, \mathbb{Z})$ regular basis	Potentially large admissible denominators	Integral on every smooth node	No known regular basis has both sufficient denominator and Chebyshev norm
Dyadic interpolation	Exact rational certificate; short proofs	Fits $2^n \mapsto 3^n$ exactly	Extrapolation denominator/error loses for large M
q -Newton	Entire geometric interpolation; explicit coefficients	Integer-valued on a geometric axis	First extrapolation is a superexponentially large integer
Discrete minimax	Exact alternation and cofactor formula	Integral evaluation matrix	Clearing coefficients changes subunit error D/S into integer error D
Rational/Padé	Can place poles near the branch set	Integer numerator/denominator at integer nodes	Locking scale is $1/ Q(q) $; coefficients often lie in $\mathbb{Q}(\beta)$
Hermite–Padé exponential	Powerful determinant and multiplicity machinery	Works for algebraic differential data	Natural arguments and system coefficients are logarithmic/transcendental
Orthogonal / Christoffel Müntz system	Exact Gram quotient; good conditioning Flexible semigroup of frequencies	Determinants integral with integral weights Every term integral on $\mathcal{B}$ under a counterexample	Real least squares ignores the evaluation-lattice coset Integer coefficient space has a hard unit floor; signed exponents incur denominators

Table 6: Approximation-theoretic approaches and their audited failure points.

The common pattern is now transparent. Real approximation can be made strong, and arithmetic

integrality can be made exact, but every known construction pays for moving from one to the other. The conjecture asks for a single auxiliary object that is simultaneously near-minimax at infinity and denominator-rich at every finite prime.

18 Conclusions

No unconditional proof of the Alaoglu–Erdős conjecture has been obtained in this continuation. The research completed here nevertheless narrows the polynomial-approximation program to a small number of exact targets.

The sharp Archimedean result is Theorem 4.3: among all multiplicative blocks, $c = 3 + 2\sqrt{2}$ maximizes the product of smooth-node density and Bernstein-ellipse rate. The optimized inequality can reach only the smallest hypothetical non-power generator $M = 3$; it is geometrically incapable of treating $M \geq 5$. This is a rigorous boundary on an entire class of classical polynomial proofs.

The sharp arithmetic result is the unit barrier. On smooth nodes, integer or conditional Müntz coefficients give integral residuals, so every unsuccessful approximation has norm at least one. The discrete cofactor formula shows exactly what happens to the real minimax solution: its subunit error D/S becomes the nonzero integer determinant D when the coefficient denominator is cleared. The missing theorem is therefore not a slightly better real approximation estimate. It is an arithmetic construction that clears the approximant without multiplying the target coefficient.

Two constructive by-products are exact. Dyadic interpolation gives short symbolic certificates excluding $M = 3, 5, 6, 7$, and the q -Newton basis gives a complete formula for integer-valued interpolation on a geometric progression, including its superexponential first-extrapolation error. These results are modest compared with the inherited finite scan, but they are independent, proof-producing, and formalization-ready.

The most promising next step is the generator-three branch in $\text{Int}(\mathcal{B}, \mathbb{Z})$. Bhargava p -orderings and their recent extensions provide a language for its allowable denominators; integer Chebyshev and potential theory provide the real norm to beat. A regular basis combining both at the optimal block would settle $3^\vartheta \notin \mathbb{Z}$. To reach all M , a stronger ingredient is needed: a weighted rational construction, a signed Müntz basis with compensated denominators, or a genuinely new logarithmic auxiliary-function theorem crossing the four-exponentials boundary.

Final status. The conjecture remains open. The continuation proves a sharp optimal-block theorem, exact polynomial and rational locking principles, a cofactor minimax identity with an arithmetic integerization theorem, a q -Newton interpolation theorem, and four symbolic generator exclusions. It also identifies the precise arithmetic approximation theorem that would turn the only favorable Bernstein branch into a proof.

A Detailed calculus for the optimal block

For reference, the ellipse parameter can be derived directly. Map $[1, c]$ affinely to $[-1, 1]$ by

$$x = \frac{2t - (1 + c)}{c - 1}.$$

The branch point $t = 0$ maps to

$$x_0 = -\frac{c+1}{c-1} < -1.$$

The Bernstein ellipse through x_0 has parameter

$$\rho = |x_0| + \sqrt{x_0^2 - 1} = \frac{c+1+2\sqrt{c}}{c-1} = \frac{\sqrt{c}+1}{\sqrt{c}-1}.$$

If $c = e^{2y}$, then

$$\rho = \frac{e^y + 1}{e^y - 1} = \coth(y/2) = e^u.$$

The half-angle identity gives

$$\tanh(y/2) = e^{-u}, \quad \sinh y = \frac{2e^{-u}}{1 - e^{-2u}} = \frac{1}{\sinh u}.$$

Thus the optimization of $\Psi(c) = 2yu$ is symmetric in y and u under the constraint $\sinh y \sinh u = 1$. Strict convexity of $z \mapsto \log \sinh(e^z)$ gives

$$\log \sinh \sqrt{yu} \leq \frac{1}{2} \log \sinh y + \frac{1}{2} \log \sinh u = 0.$$

Therefore $yu \leq \operatorname{arsinh}(1)^2$, with equality only at $y = u = \operatorname{arsinh}(1)$.

The numerical constants are

$$\begin{aligned} c_* &= 5.8284271247461900976\dots, \\ \rho(c_*) &= 2.4142135623730950488\dots, \\ \max \Psi &= 1.5536387997913919630\dots, \\ M_{\text{crit}} &= 4.1131245860194105174\dots \end{aligned}$$

B Proof details for the generalized power zero bound

Let $\lambda_0 < \dots < \lambda_s$ be distinct real numbers. The Wronskian of $t^{\lambda_0}, \dots, t^{\lambda_s}$ on $(0, \infty)$ is

$$W(t) = t^{\lambda_0 + \dots + \lambda_s - s(s+1)/2} \prod_{0 \leq i < j \leq s} (\lambda_j - \lambda_i).$$

It never vanishes. Hence the functions form an extended complete Chebyshev system, and any nontrivial linear combination has at most s positive zeros counted with multiplicity. This proves every zero-count statement used in the polynomial, rational, and Müntz locking theorems.

An elementary induction is also available. Divide a nonzero generalized polynomial by its smallest power, differentiate, and apply Rolle's theorem. Each differentiation removes the constant term and leaves one fewer exponent. This proves the distinct-zero form without the full Wronskian formalism.

C Derivation of the discrete cofactor formula

Let A be an $(N+1) \times N$ real matrix of rank N , $f \in \mathbb{R}^{N+1}$, and $V = [A \ f]$. For each i , let

$$C_i = (-1)^{i+N} \det A_{\hat{i}},$$

where $A_{\hat{i}}$ deletes row i . Laplace expansion gives

$$A^T C = 0, \quad C^T f = \det V = D.$$

The dual of the distance from f to $\text{im } A$ in ℓ^∞ is

$$\max\{y^T f : A^T y = 0, \|y\|_1 \leq 1\}.$$

The nullspace is spanned by C , so the value is $|D|/S$ with $S = \sum_i |C_i|$. If the ordered evaluation matrix is strictly totally positive, then the C_i alternate in sign. Setting $s_i = \text{sgn}(C_i)$, the square matrix $B = [A \ s]$ has

$$\det B = C^T s = S.$$

The equation

$$Ac + \frac{D}{S}s = f$$

therefore has a unique solution and is the alternant. If A, f are integral, multiply by S :

$$B \begin{pmatrix} Sc \\ D \end{pmatrix} = Sf.$$

Since $\text{adj}(B)$ is integral,

$$\begin{pmatrix} Sc \\ D \end{pmatrix} = \text{adj}(B)f \in \mathbb{Z}^{N+1}.$$

This proves Theorems 7.1 and 7.2.

D The geometric Newton identity

For $q \geq 2$ and $n \geq k$, one has

$$B_k^{(q)}(q^n) = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i} = \prod_{i=0}^{k-1} \frac{q^{n-i} - 1}{q^{k-i} - 1} = \left[\begin{matrix} n \\ k \end{matrix} \right]_q.$$

For $n < k$, the product vanishes because one factor is $q^n - q^n$. The Newton coefficient at stage k is the error at the new node q^k , hence

$$a_k = r^k - P_{k-1}(q^k).$$

Induction using the Gaussian-binomial recursion gives

$$a_k = \prod_{i=0}^{k-1} (r - q^i).$$

Equivalently, the exact identity

$$r^n = \sum_{k=0}^n \left[\begin{matrix} n \\ k \end{matrix} \right]_q \prod_{i=0}^{k-1} (r - q^i)$$

is the Newton expansion of the data $q^n \mapsto r^n$. Truncating at d gives Theorem 8.1; at $n = d+1$ only one tail term survives.

E Literature chronology and relevance matrix

Date	Source	Relevance to this continuation
1944	Alaoglu–Erdős	Original arithmetic question in the theory of highly composite numbers.
1957	Schneider	Places the surrounding four-exponentials problem among central open transcendence questions.
1997	Bhargava	p -orderings and generalized factorials for integer-valued polynomial rings on arbitrary subsets.
2003	Borwein–Pinner–Pritsker	Monic integer Chebyshev problem and exact constants on arithmetic compact sets.
2005	Pritsker	Integer Chebyshev polynomials, potential theory, capacity, and small integer-coefficient polynomials.
2023	Waldschmidt	Modern survey explicitly connecting the question to four exponentials and Schanuel-type conjectures.
2024/2026	Dasgupta–Kakde	Matrix Coefficient Conjecture and its relation to strong four exponentials; published in <i>Advances in Mathematics</i> .
2025	Massold	Transcendence-degree estimates for exponentials of products; stronger cases reduced to torus intersections.
2025	Fischler–Rivoal	New exponential-value transcendence measure using Hermite–Padé approximants and an optimized Siegel determinant.
2025	Adamczewski–Faverjon	Explicit algebraic-independence measures for values of E - and M -functions at algebraic points.
2025/2026	Gorbachev–Ivanov–Tikhonov	Discrete Chebyshev alternation and Sturm oscillation for discrete polynomial systems.
2025/2026	Berger–Sprang	Links regular bases of integer-valued polynomial rings with p -adic Fourier theory.
2025	Lagarias–Yangjit	Extends Bhargava-style orderings and generalized factorials from prime ideals to general ideals.

Table 7: Selected literature used to delimit the approximation program.

The chronology shows that the relevant tools are progressing on both sides: quantitative auxiliary functions at the Archimedean place and generalized integer-valued polynomial bases at finite places. What remains absent is a theorem joining them for the specialized 2×2 logarithmic matrix.

References

- [1] L. Alaoglu and P. Erdős, *On highly composite and similar numbers*, Transactions of the American Mathematical Society **56** (1944), 448–469. DOI: [10.1090/S0002-9947-1944-0011087-2](https://doi.org/10.1090/S0002-9947-1944-0011087-2).
- [2] Th. Schneider, *Einführung in die transzendenten Zahlen*, Springer, 1957.
- [3] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Grundlehren der mathematischen Wissenschaften 326, Springer, 2000.
- [4] M. Waldschmidt, *The Four Exponentials Problem and Schanuel’s Conjecture*, in J.-M. Morel and B. Teissier (eds.), *Mathematics Going Forward*, Lecture Notes in Mathematics 2313, Springer, 2023, pp. 579–592. DOI: [10.1007/978-3-031-12244-6_39](https://doi.org/10.1007/978-3-031-12244-6_39).
- [5] S. Dasgupta and M. Kakde, *Ranks of Matrices of Logarithms of Algebraic Numbers II: The Matrix Coefficient Conjecture*, arXiv:2408.08178, 2024; subsequently published in 2026 as cited below.
- [6] S. Dasgupta and M. Kakde, *Ranks of matrices of logarithms of algebraic numbers II: The Matrix Coefficient Conjecture*, Advances in Mathematics **487** (2026), Article 110753. DOI: [10.1016/j.aim.2025.110753](https://doi.org/10.1016/j.aim.2025.110753).
- [7] H. Massold, *Transcendence degrees of fields generated by exponentials of products*, arXiv:2506.01123, 2025.
- [8] S. Fischler and T. Rivoal, *A new transcendence measure for the values of the exponential function at algebraic arguments*, arXiv:2502.17992v1, February 2025.
- [9] C. Faverjon and B. Adamczewski, *Algebraic Independence Measures for Values of E-functions and M-functions*, arXiv:2502.09999, 2025.
- [10] D. V. Gorbachev, V. I. Ivanov, and S. Yu. Tikhonov, *Chebyshev systems and Sturm oscillation theory for discrete polynomials*, Forum of Mathematics, Sigma **14** (2026), e84. DOI: [10.1017/fms.2026.10168](https://doi.org/10.1017/fms.2026.10168); arXiv:2501.02358v2, August 2026.
- [11] M. Bhargava, *P-orderings and polynomial functions on arbitrary subsets of Dedekind rings*, Journal für die reine und angewandte Mathematik **490** (1997), 101–127.
- [12] J. C. Lagarias and W. Yangjit, *B-orderings for all ideals B of Dedekind domains and generalized factorials*, arXiv:2502.19072, 2025.
- [13] L. Berger and J. Sprang, *Integer-valued polynomials and p-adic Fourier theory*, Rendiconti del Seminario Matematico della Università di Padova, Online First, July 21, 2026. DOI: [10.4171/RSMUP/204](https://doi.org/10.4171/RSMUP/204); arXiv:2502.18053v2, April 2025.
- [14] I. E. Pritsker, *Small polynomials with integer coefficients*, Journal d’Analyse Mathématique **96** (2005), 151–190. DOI: [10.1007/BF02787827](https://doi.org/10.1007/BF02787827).
- [15] P. B. Borwein, C. G. Pinner, and I. E. Pritsker, *Monic integer Chebyshev problem*, Mathematics of Computation **72** (2003), no. 244, 1901–1916. DOI: [10.1090/S0025-5718-03-01477-7](https://doi.org/10.1090/S0025-5718-03-01477-7).
- [16] S. Karlin and W. J. Studden, *Tchebycheff Systems: With Applications in Analysis and Statistics*, Interscience, 1966.

- [17] G. G. Lorentz, *Approximation of Functions*, second edition, Chelsea, 1986.
- [18] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, 2013.
- [19] P.-J. Cahen and J.-L. Chabert, *Integer-Valued Polynomials*, Mathematical Surveys and Monographs 48, American Mathematical Society, 1997.
- [20] The ProveIt project, *The Alaoglu–Erdős Integer-Exponent Conjecture: Machine-Checked Partial Results, Exact Conditional Structure, Determinant Methods, and the Remaining Barrier*, supplied unified report, August 21, 2026.